On JEPA Isotropy

Anonymous Author(s)
Affiliation
Address
`email`

Abstract

We study the empirical risk minimization (ERM) of linear Joint Embedding Predictive Architecture (JEPA) and establish its key conditions for optimality. By casting the embedding bottleneck as a rank constraint, we formulate JEPA as a reduced-rank regression problem. This induces a risk decomposition into an irreducible error and an excess-risk bound. Within the irreducible term, we find the rank of the optimal predictor saturates with the quality of the target. Within the excess-risk bound, we find spectral conditioning of the whitened end-to-end predictor governs minimization. Together, they prescribe an isotropic regularization procedure. Under this regularization, we show such JEPA isotropy subsumes canonical JEPA isotropic embedding deigns and hence justifies their empirical successes via ERM. Numerical validations corroborate our theory.

1 Introduction

We study the finite-sample risk of Joint-Embedding Predictive Architecture (JEPA) [1, 2] and establish its key conditions for optimality. JEPA has recently become a self-supervised representation learning paradigm [2–5] by jointly training a context encoder, a target encoder, and a predictor so that the predicted target embedding matches the target's.

A key recipe in JEPA training is to enforce isotropy on the encoder's marginal embedding, either through architecture [2–4] or distributional regularization [6, 5]. Their justifications for this recipe are partial: heuristic by analogy to non-predictive self-supervised learning family [7–9], passive in preventing failure modes such as representational [10, 11] or early training collapse [12], or theoretically scoped to specific probes or data assumptions [5, 13]. More fundamentally, these designs regularize only the encoder and ignore the predictor, the defining component of JEPA beyond plain Joint-Embedding Architectures (JEAs). We resolve this through a comprehensive finite-sample analysis of JEPA that jointly considers its encoder and predictor, revealing an end-to-end predictive isotropy condition that subsumes and justifies these encoder-only designs above.

Casting the embedding bottleneck as a rank constraint, we formulate multi-target JEPA as reduced-rank regression [14, 15]. This induces a decomposition of the empirical risk into an irreducible error and a finite-sample excess-risk bound (Proposition 3.20). Within the irreducible term, we identify two laws governing the rank of the optimal predictor. It grows with the number of targets K only when new targets contribute non-redundant directions, a heterogeneity condition (Theorem 3.5), and saturates at the embedding bottleneck d (Proposition 3.9). Any non-redundant directions beyond d contribute an irreducible error. Within the excess-risk bound, we show the spectral conditioning of the whitened end-to-end predictor T_K governs its minimization. The bound is uniquely minimized when the eigenvalues of $H_K := T_K^\top T_K$ are flat, a condition we call predictive isotropy (Lemma 3.22).

Together, the two terms prescribe an isotropic regularization on the end-to-end prediction. The irreducible term motivates target heterogeneity to fill the bottleneck; the excess-risk bound motivates flattening the spectrum of H_K. Both prescriptions reduce to regularizing T_K's spectrum toward predictive isotropy. Under this regularization, we show predictive isotropy on T_K subsumes the

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

canonical isotropic-embedding paradigm. Through a gauge-invariant factorization of the rank-d optimum, the population-optimal encoder under this regularization automatically realizes the isotropic-Gaussian embedding that prior paradigms enforce externally (Lemma 3.26 and Theorems 3.27 and 3.28). Encoder-marginal isotropy thus emerges as a corollary of empirical risk minimization, justifying the empirical success of canonical isotropic embeddings.

Numerical experiments in a controlled linear-Gaussian setting verify each predicted phenomenon: rank growth, heterogeneity-driven recovery, the bottleneck capacity floor, and the conditioning behavior at predictive isotropy.

Contribution.

- Reduced-rank regression formulation and risk decomposition.
- Rank growth and heterogeneity in the irreducible term.
- Predictive isotropy as the excess-risk minimizer.
- A maximum-entropy regularizer on the end-to-end prediction. Isotropic-Gaussian embedding emerges as a corollary.

2 Preliminaries

In this section, we introduce the notation and background needed for our analysis. We first present the Joint-Embedding Predictive Architecture (JEPA) with frozen-teacher and linear assumptions (Section 2.1), then review reduced-rank regression (Section 2.2).

2.1 JEPA with Frozen Teacher

Let $z \in \mathbb{R}^D$ denote a data sample drawn from a distribution $\mathcal{P}$ on $\mathbb{R}^D$. A *mask* is an index set $m \subseteq [D]$ selecting a subset of coordinates. A *mask distribution* $\mathcal{D}$ is a probability distribution over tuples $(m_c, m_{t,1}, \ldots, m_{t,K})$ specifying one *context block* and $K \geq 1$ *target blocks*.[1] For a sampled tuple, write $z_{\text{ctx}} := z_{m_c} \in \mathbb{R}^{D_x}$ and $z_{\text{tgt}}^{(k)} := z_{m_{t,k}} \in \mathbb{R}^{D_y^{(k)}}$ for the corresponding context and target subvectors. JEPA jointly trains an encoder and a predictor to predict the embedding of each target block from the embedding of the context block.

Architecture. JEPA consists of three components:

(i) A *context encoder* $f_\theta : \mathbb{R}^{D_x} \to \mathbb{R}^d$ mapping the context block to an embedding $x = f_\theta(z_{\text{ctx}})$;

(ii) A *target encoder* (*teacher*) $f_{\bar\theta} : \mathbb{R}^{D_y^{(k)}} \to \mathbb{R}^d$ mapping each target to an embedding $y^{(k)} = f_{\bar\theta}(z_{\text{tgt}}^{(k)})$;

(iii) A *predictor* $g_W : \mathbb{R}^d \times \mathcal{Z} \to \mathbb{R}^d$ that predicts the k-th target embedding $y^{(k)}$ from the context embedding x and target-side metadata $q_k \in \mathcal{Z}$, i.e., $\widehat{y}^{(k)} = g_W(x, q_k)$.

Throughout this work, we assume a frozen teacher and a linear end-to-end map.

Assumption 2.1 (Frozen Teacher)**.** The target encoder $f_{\bar\theta}$ is fixed during training of the context encoder f_θ and predictor g_W.

Assumption 2.2 (End-to-end linearity)**.** The end-to-end map $g_W \circ f_\theta : \mathbb{R}^{D_x} \to \mathbb{R}^{Kd}$ is linear.

For explicit matrix-level analysis, we further specialize to a specific linear factorization.

Assumption 2.3 (Linear factorization)**.** The context encoder is linear, and for each target block $k \in [K]$ with fixed metadata q_k, the predictor is linear in the context embedding:

$$f_\theta(z_{\text{ctx}}) = A z_{\text{ctx}}, \quad g_W(x, q_k) = W_k x, \qquad A \in \mathbb{R}^{d \times D_x}, \quad W_k \in \mathbb{R}^{d \times d}.$$

Assumption 2.3 implies Assumption 2.2 and serves as the working factorization for Section 3.1-3.5. Section 3.6 relaxes it to nonlinear factorizations within the same end-to-end linear map (Lemma 3.26).

Under Assumption 2.3, stacking the target embeddings,

$$Y := \begin{bmatrix} (y^{(1)})^\top & \cdots & (y^{(K)})^\top \end{bmatrix}^\top \in \mathbb{R}^{Kd}, \tag{1}$$

$$\widehat{Y} = \begin{bmatrix} \widehat{y}^{(1)} \\ \vdots \\ \widehat{y}^{(K)} \end{bmatrix} = Wx, \qquad W := \begin{bmatrix} W_1 \\ \vdots \\ W_K \end{bmatrix} \in \mathbb{R}^{Kd \times d}. \tag{2}$$

Thus the end-to-end map from context to predicted target embeddings is $WA \in \mathbb{R}^{Kd \times D_x}$.

[1] We call a mask a *block mask* when it selects a contiguous index set. The target blocks need not be disjoint.

Below we define the JEPA training objective. Unless otherwise stated, all expectations are taken over $z \sim \mathcal{P}$ and $(m_c, m_{t,1}, \ldots, m_{t,K}) \sim \mathcal{D}$.

Definition 2.4 (JEPA loss)**.** Let $x := Az_{\mathrm{ctx}} \in \mathbb{R}^d$ and stack the target embeddings $y^{(k)} := f_{\bar{\theta}}(z_{\mathrm{tgt}}^{(k)})$ into Y as in (1), and the predictor blocks W_k into W as in (2). The K-target JEPA loss is

$$\mathcal{L}_K(W, A) \;:=\; \frac{1}{K}\,\mathbb{E}\big[\|Y - Wx\|^2\big]. \tag{3}$$

The composition WA acts on the raw context x only through its row space, a low-dimensional subspace of $\mathbb{R}^{D_x}$ that captures the directions JEPA effectively operates on. This subspace is central to our analysis, so we name it next.

Definition 2.5 (Predictive subspace)**.** A linear subspace $V \subseteq \mathbb{R}^{D_x}$ is called a *predictive subspace* if $V = \mathrm{row}(WA)$ for some admissible predictor WA under Assumption 2.3. We write $\mathcal{V}_r$ for the set of predictive subspaces of dimension r.

Denote the population second moments $\Sigma_{XX} := \mathbb{E}[xx^\top] \in \mathbb{R}^{d\times d}$, $\Sigma_{Y^{(k)}X} := \mathbb{E}[y^{(k)}x^\top] \in \mathbb{R}^{d\times d}$ for $k \in [K]$, and their stack $\Sigma_{YX} := \mathbb{E}[Yx^\top] = [\Sigma_{Y^{(1)}X}^\top, \ldots, \Sigma_{Y^{(K)}X}^\top]^\top \in \mathbb{R}^{Kd\times d}$. Assume $\Sigma_{XX} \succ 0$ (non-degenerate context embedding).

2.2 Reduced-Rank Regression

Given paired random vectors $x \in \mathbb{R}^p$ and $y \in \mathbb{R}^q$, the population squared-loss regression problem

$$\min_{B\in\mathbb{R}^{q\times p}} \mathbb{E}\big[\|y - Bx\|^2\big] \tag{4}$$

admits the closed-form ordinary least squares (OLS) minimizer $B_{\mathrm{OLS}} := \Sigma_{YX}\Sigma_{XX}^{-1}$ when $\Sigma_{XX} \succ 0$.

Reduced-rank regression (RRR) [15, 16] restricts the minimization in (4) to $\mathrm{rank}(B) \le r$.

Definition 2.6 (Reduced-rank regression problem)**.** Given paired $x \in \mathbb{R}^p$, $y \in \mathbb{R}^q$ as predictor and response random vectors, for a prescribed rank $r \le \min(p, q)$, the reduced-rank regression is

$$B_{\mathrm{RRR}} \in \arg\min_{B\in\mathbb{R}^{q\times p}} \mathbb{E}\big[\|y - Bx\|^2\big] \quad \text{subject to} \quad \mathrm{rank}(B) \le r. \tag{5}$$

The constraint $\mathrm{rank}(B) \le r$ means the linear predictor factors through an r-dimensional latent subspace. The following lemma gives a closed-form solution.

Lemma 2.7 (RRR solution, Theorem 2.2 of [16])**.** *Let $M := \Sigma_{XX}^{1/2}B_{\mathrm{OLS}}^\top B_{\mathrm{OLS}}\Sigma_{XX}^{1/2}$, and let $V_r \in \mathbb{R}^{p\times r}$ collect eigenvectors corresponding to the top r eigenvalues of M. A solution to (5) is*

$$B_{\mathrm{RRR}} = B_{\mathrm{OLS}}\Sigma_{XX}^{1/2}V_rV_r^\top\Sigma_{XX}^{-1/2}. \tag{6}$$

Equivalently, RRR solution is obtained by taking the best rank-r approximation of $B_{\mathrm{OLS}}\Sigma_{XX}^{1/2}$ and then unwhitening by $\Sigma_{XX}^{-1/2}$. Informally, this means that RRR retains the r predictor-space directions that are most useful for linearly predicting y.

Remark 2.8 (Connection to CCA and PCA). Reduced-rank regression is closely related to canonical correlation analysis (CCA): after appropriate whitening, the rank-r solution retains the leading canonical predictive directions linking the predictor space to the response space. As a special case, when $y = x$, the corresponding reduced-rank self-prediction problem recovers the principal subspace of Σ_{XX}, linking RRR to PCA.

3 Methodology

3.1 JEPA as Reduced-Rank Regression

We show that, under the frozen-teacher and linearity assumptions (Assumptions 2.1 and 2.3), the JEPA objective reduces to a reduced-rank regression problem.

For a prescribed rank $r \le d$, consider the rank-constrained JEPA problem

$$W_r^\star \in \arg\min_{W\in\mathbb{R}^{Kd\times d}} \mathcal{L}_K(W, A) \qquad \text{subject to} \qquad \mathrm{rank}(W) \le r, \tag{7}$$

where $\mathcal{L}_K(W, A)$ is the JEPA loss in Definition 2.4.

Proposition 3.1 (JEPA as reduced-rank regression)**.** *Under Assumptions 2.1 and 2.3, the optimization problem (7) is exactly the reduced-rank regression problem in Definition 2.6 with $p = d$, $q = Kd$, predictor $x \in \mathbb{R}^d$, and response $Y \in \mathbb{R}^{Kd}$.*

Proof. From Definition 2.4,

$$\mathcal{L}_K(W, A) = \frac{1}{K}\,\mathbb{E}\big[\|Y - Wx\|^2\big].$$

Since the factor $1/K$ does not affect the minimizer, (7) is equivalent to

$$W_r^\star \in \arg\min_{W\in\mathbb{R}^{Kd\times d}} \mathbb{E}\big[\|Y - Wx\|^2\big] \qquad \text{subject to} \qquad \operatorname{rank}(W) \le r.$$

This is precisely the reduced-rank regression problem in Definition 2.6. $\square$

Let $B_{\mathrm{OLS}} \coloneqq \Sigma_{YX}\Sigma_{XX}^{-1}$, $M \coloneqq \Sigma_{XX}^{1/2} B_{\mathrm{OLS}}^\top B_{\mathrm{OLS}} \Sigma_{XX}^{1/2}$, and let $V_r \in \mathbb{R}^{d\times r}$ collect the eigenvectors corresponding to the top r eigenvalues of M. The closed-form solution of JEPA problem (7) therefore follows from Lemma 2.7.

Corollary 3.2 (Closed-form JEPA reduced-rank solution)**.** *Under the assumptions of Proposition 3.1, a solution to* (7) *is*

$$W_r^\star = B_{\mathrm{OLS}}\Sigma_{XX}^{1/2} V_r V_r^\top \Sigma_{XX}^{-1/2}.$$

Thus, JEPA inherits the classical reduced-rank structure of multivariate regression, with predictor dimension d and response dimension Kd. In particular, the solution $W_r^\star$ is obtained by truncating the unconstrained regression map $\Sigma_{YX}\Sigma_{XX}^{-1}$ to its top-r canonical predictive directions.

3.2 Multi-Target Masking and Rank Growth

This subsection studies when increasing the number of target blocks can enlarge the predictive structure accessible from the context embedding x. The JEPA predictor in Equation (7) is subject to two rank bottlenecks: the explicit constraint r, and an intrinsic bottleneck induced by the data distribution through x. Lemma 3.3 makes the latter precise:

Lemma 3.3 (Rank of the reduced-rank JEPA solution)**.** *Let $W_r^\star$ be the closed-form solution from Corollary 3.2. Then*

$$\operatorname{rank}(W_r^\star) = \min\{r,\, \operatorname{rank}(\Sigma_{YX})\}. \tag{8}$$

Proof. Using Corollary 3.2 and $B_{\mathrm{OLS}} = \Sigma_{YX}\Sigma_{XX}^{-1}$,

$$W_r^\star = \Sigma_{YX}\Sigma_{XX}^{-1/2}\, V_r V_r^\top\, \Sigma_{XX}^{-1/2}.$$

Since $\Sigma_{XX} \succ 0$, the factor $\Sigma_{XX}^{-1/2}$ is invertible, so

$$\operatorname{rank}(W_r^\star) = \operatorname{rank}\Big(\Sigma_{YX}\Sigma_{XX}^{-1/2}\, V_r V_r^\top\Big).$$

Let $\Sigma_{YX}\Sigma_{XX}^{-1/2} = U\, \operatorname{diag}(\sigma_1, \dots, \sigma_{\operatorname{rank}(\Sigma_{YX})}, 0, \dots, 0)\, V^\top$ be an SVD. By the identity

$$M = \Sigma_{XX}^{1/2} B_{\mathrm{OLS}}^\top B_{\mathrm{OLS}} \Sigma_{XX}^{1/2} = \big(\Sigma_{YX}\Sigma_{XX}^{-1/2}\big)^\top \big(\Sigma_{YX}\Sigma_{XX}^{-1/2}\big),$$

the columns of V_r are the top-r right singular vectors of $\Sigma_{YX}\Sigma_{XX}^{-1/2}$. Since V is orthogonal, $V_r V_r^\top$ projects onto the top-r right-singular subspace, so

$$\Sigma_{YX}\Sigma_{XX}^{-1/2}\, V_r V_r^\top = U\, \operatorname{diag}(\sigma_1, \dots, \sigma_{\min(r,\, \operatorname{rank}(\Sigma_{YX}))}, 0, \dots, 0)\, V^\top,$$

which has rank $\min\{r,\, \operatorname{rank}(\Sigma_{YX})\}$. Using the fact that $\operatorname{rank}(\Sigma_{YX}\Sigma_{XX}^{-1/2}) = \operatorname{rank}(\Sigma_{YX})$ (again because $\Sigma_{XX}^{-1/2}$ is invertible) yields (8). $\square$

Lemma 3.3 identifies $\operatorname{rank}(\Sigma_{YX})$ as the intrinsic bottleneck predictive rank: adding target blocks enlarges the predictive structure only insofar as it raises this rank. To study how $\operatorname{rank}(\Sigma_{YX})$ depends on the choice and number of target blocks, we aggregate across blocks. Recall $\Sigma_{Y^{(k)}X}, \Sigma_{YX}$ defined in Section 2.1. It is natural to define the predictor-side Gram matrix

$$G_K \coloneqq \Sigma_{YX}^\top \Sigma_{YX} = \sum_{k=1}^{K} \Sigma_{Y^{(k)}X}^\top \Sigma_{Y^{(k)}X} \in \mathbb{R}^{d\times d}. \tag{9}$$

Each summand $\Sigma_{Y^{(k)}X}^\top \Sigma_{Y^{(k)}X}$ is positive semidefinite, so adding a block can only enlarge the range of G_K (equivalently, shrink its kernel). Throughout, we write $\lambda_i(\cdot)$ and $\sigma_i(\cdot)$ for the i-th largest eigenvalue and singular value (in decreasing order), respectively, and $\|\cdot\|_{\mathrm{op}}$ for the operator norm. Since $G_K = \Sigma_{YX}^\top \Sigma_{YX}$, its eigenvalues coincide with squared singular values of Σ_{YX}; in particular, $\lambda_i(G_K) = \sigma_i(\Sigma_{YX})^2$ for every i, and that

$$\lambda_1(G_K) = \sigma_1(\Sigma_{YX})^2 = \|\Sigma_{YX}\|_{\mathrm{op}}^2. \tag{10}$$

Definition 3.4 (Predictive rank)**.** For a given number of target blocks K, the *predictive rank* is

$$r_\star(K) \coloneqq \operatorname{rank}(\Sigma_{YX}).$$

Since $\Sigma_{XX} \succ 0$ and $G_K = \Sigma_{YX}^\top \Sigma_{YX}$, one has the equivalent forms $r_\star(K) = \operatorname{rank}(G_K) = \operatorname{rank}(B_{\mathrm{OLS}})$, where $B_{\mathrm{OLS}} = \Sigma_{YX}\Sigma_{XX}^{-1}$; and since $G_K \in \mathbb{R}^{d\times d}$, $r_\star(K) \le d$. Intuitively, $r_\star(K)$ is the dimension of a predictive subspace (Definition 2.5). Here we study how it depends on K.

Theorem 3.5 (Rank growth)**.** *Let $G_K \coloneqq \Sigma_{YX}^\top \Sigma_{YX}$ and $r_\star(K) \coloneqq \operatorname{rank}(\Sigma_{YX})$ for $K \ge 1$. Then:*

1. ***Monotonicity.***
$$r_\star(K+1) \ge r_\star(K).$$
2. ***Strict growth under complementarity.*** *If $\ker(G_K) \not\subseteq \ker(\Sigma_{Y^{(K+1)}X})$, then*
$$r_\star(K+1) > r_\star(K).$$

Proof. From (9),

$$G_{K+1} = G_K + \Sigma_{Y^{(K+1)}X}^\top \Sigma_{Y^{(K+1)}X}, \tag{11}$$

so for every $v \in \mathbb{R}^d$,

$$v^\top G_{K+1} v = v^\top G_K v + \|\Sigma_{Y^{(K+1)}X}\, v\|^2.$$

Both terms on the right are nonnegative, so $v^\top G_{K+1} v = 0$ holds if and only if both $v^\top G_K v = 0$ and $\Sigma_{Y^{(K+1)}X} v = 0$. Hence

$$\ker(G_{K+1}) = \ker(G_K) \cap \ker(\Sigma_{Y^{(K+1)}X}) \subseteq \ker(G_K).$$

Therefore

$$\dim\ker(G_{K+1}) \le \dim\ker(G_K).$$

Since $G_K \in \mathbb{R}^{d\times d}$, rank-nullity gives

$$\operatorname{rank}(G_{K+1}) = d - \dim\ker(G_{K+1}) \ge d - \dim\ker(G_K) = \operatorname{rank}(G_K).$$

Using $\operatorname{rank}(G_K) = \operatorname{rank}(\Sigma_{YX}) = r_\star(K)$, we obtain $r_\star(K+1) \ge r_\star(K)$, proving (1).

For (2), suppose $\ker(G_K) \not\subseteq \ker(\Sigma_{Y^{(K+1)}X})$ and pick $v \in \ker(G_K) \setminus \ker(\Sigma_{Y^{(K+1)}X})$. Then $v \in \ker(G_K)$ but $v \notin \ker(G_{K+1})$, so the inclusion above is strict, giving $r_\star(K+1) > r_\star(K)$. □

Corollary 3.6 (Saturation)**.** *There exists $K^\star$ such that $r_\star(K) = r_\star(K^\star)$ for every $K \ge K^\star$.*

Proof. By Theorem 3.5 (1), $r_\star(\cdot)$ is monotone. Since $G_K \in \mathbb{R}^{d\times d}$, we have $r_\star(K) = \operatorname{rank}(G_K) \le d$, so $r_\star(K)$ is a nondecreasing integer sequence bounded by d, hence eventually constant. □

Beyond the rank itself, it will be useful to track how far G_K is from being rank-deficient on its range, i.e., its smallest nonzero eigenvalue.

Definition 3.7 (Heterogeneity level)**.** The heterogeneity level of the family $\{\Sigma_{Y^{(k)}X}\}_{k=1}^K$ is

$$\lambda_{\mathrm{het}} \coloneqq \lambda_{r_\star(K)}(G_K), \tag{12}$$

where $\lambda_{r_\star(K)}(G_K)$ denotes the $r_\star(K)$-th largest eigenvalue of G_K.

Since $G_K \succeq 0$, its eigenvalues are nonnegative, and since $\operatorname{rank}(G_K) = r_\star(K)$, exactly $r_\star(K)$ of them are nonzero. Therefore, $\lambda_{r_\star(K)}(G_K)$ is precisely the smallest nonzero eigenvalue of G_K. By the identity $G_K = \Sigma_{YX}^\top \Sigma_{YX}$, Definition 3.7 is equivalent to a singular-value floor:

$$\sigma_{r_\star(K)}(\Sigma_{YX}) = \sqrt{\lambda_{\mathrm{het}}}. \tag{13}$$

Remark 3.8 (Heterogeneity as complementarity). Theorem 3.5 shows that adding target blocks does not by itself create new predictive directions. Rank increases only if the new block cross-covariance $\Sigma_{Y^{(K+1)}X}$ is nonzero on $\ker(G_K)$, that is, only if it reveals directions of X that were invisible to the previously accumulated structure. Equivalently, strict growth fails when $\Sigma_{Y^{(K+1)}X}$ vanishes on $\ker(G_K)$, as in the case of highly overlapping or redundant target blocks. In this sense, heterogeneity means complementarity among blockwise cross-covariances, not merely a larger number of targets.

By (13), a larger heterogeneity level λ_{het} yields a larger smallest nonzero singular value of the stacked cross-covariance Σ_{YX}. The next proposition shows that this improves finite-sample recovery of its top-$r_\star(K)$ right singular subspace, which is a subspace of the embedding space $\mathbb{R}^d$.

Proposition 3.9 (Finite-sample subspace recovery under heterogeneity)**.** *Assume the family $\{\Sigma_{Y^{(k)}X}\}_{k=1}^K$ has heterogeneity level $\lambda_{\mathrm{het}} > 0$ (Definition 3.7), and let $\widehat{\Sigma}_{YX}$ be the empirical cross-covariance computed from n i.i.d. samples under the sub-Gaussian assumptions of Lemma A.1.*

Let $V_{r_\star(K)}, \widehat{V}_{r_\star(K)}$ denote the top-$r_\star(K)$ right singular subspaces of Σ_{YX} and $\widehat{\Sigma}_{YX}$ respectively. Then for any $\delta \in (0,1)$, with probability at least $1-\delta$,

$$\|\sin\Theta(V_{r_\star(K)}, \widehat{V}_{r_\star(K)})\|_{\mathrm{op}} \leq \frac{\epsilon_n}{\sqrt{\lambda_{\mathrm{het}}} - \epsilon_n}, \quad \epsilon_n := C\|\Sigma_{YX}\|_{\mathrm{op}}\sqrt{\tfrac{\mathrm{effdim}(\Sigma_{YX}) + \log(1/\delta)}{n}},$$

whenever $\epsilon_n < \sqrt{\lambda_{\mathrm{het}}}$.

Proof. Lemma A.1 gives $\|\widehat{\Sigma}_{YX} - \Sigma_{YX}\|_{\mathrm{op}} \leq \epsilon_n$ with probability $1-\delta$. Since $\mathrm{rank}(\Sigma_{YX}) = r_\star(K)$, applying Lemma A.3 with $r = r_\star(K)$, $\epsilon = \epsilon_n$, and Wedin gap $g_{r_\star}(\Sigma_{YX}) = \sqrt{\lambda_{\mathrm{het}}}$ (13) yields the claim. □

For sufficiently large n such that $\epsilon_n \leq \sqrt{\lambda_{\mathrm{het}}}/2$, the bound simplifies to

$$\|\sin\Theta(V_{r_\star(K)}, \widehat{V}_{r_\star(K)})\|_{\mathrm{op}} \leq \frac{2\epsilon_n}{\sqrt{\lambda_{\mathrm{het}}}} = \widetilde{O}\left(\frac{1}{\sqrt{n\lambda_{\mathrm{het}}}}\right).$$

Thus, stronger heterogeneity (larger λ_{het}) tightens the $\sin\Theta$ bound and yields more stable finite-sample recovery, and hence better identifiability, of the $r_\star(K)$ right singular subspace of Σ_{YX} in the embedding space. Conversely, redundant target blocks shrink λ_{het} and destabilize recovery, consistent with Remark 3.8.

3.3 Capacity Limits of the Embedding Bottleneck

This subsection studies the architectural bottleneck itself. Since JEPA predicts through a d-dimensional representation $x = Az_{\mathrm{ctx}}$, its end-to-end predictor from the raw context block to the stacked target embedding must factor through a rank at most d map. This induces an approximation limit even when the raw context contains richer predictive structure. For convenience, we denote

$$Z := z_{\mathrm{ctx}} \in \mathbb{R}^{D_x}, \qquad \Sigma_{ZZ} := \mathbb{E}[ZZ^\top], \qquad \Sigma_{YZ} := \mathbb{E}[YZ^\top],$$

and assume $\Sigma_{ZZ} \succ 0$. Define the unconstrained population linear predictor from raw context to stacked targets by $B^{\mathrm{raw}}_{\mathrm{OLS}} := \Sigma_{YZ}\Sigma_{ZZ}^{-1} \in \mathbb{R}^{Kd\times D_x}$. The next result formalizes the irreducible approximation error imposed by the d-dimensional embedding bottleneck.

Proposition 3.10 (Capacity lower bound from the embedding bottleneck)**.** *Let $T_K := B^{\mathrm{raw}}_{\mathrm{OLS}}\Sigma_{ZZ}^{1/2}$ and $\sigma_1(T_K) \geq \sigma_2(T_K) \geq \cdots \geq 0$ denote its singular values. Then for every $(W, A) \in \mathbb{R}^{Kd\times d} \times \mathbb{R}^{d\times D_x}$,*

$$\mathbb{E}\big[\|Y - WAZ\|_2^2\big] \geq \mathbb{E}\big[\|Y - B^{\mathrm{raw}}_{\mathrm{OLS}}Z\|_2^2\big] + \sum_{j>d}\sigma_j(T_K)^2. \tag{14}$$

Proof. For any pair (W, A), define the end-to-end map

$$B := WA \in \mathbb{R}^{Kd\times D_x}.$$

Since $W \in \mathbb{R}^{Kd\times d}$ and $A \in \mathbb{R}^{d\times D_x}$, one has $\mathrm{rank}(B) \leq d$. Conversely, every matrix $B \in \mathbb{R}^{Kd\times D_x}$ with $\mathrm{rank}(B) \leq d$ admits a factorization $B = WA$ for some such (W, A). Hence

$$\inf_{W,A}\mathbb{E}\big[\|Y - WAZ\|_2^2\big] = \inf_{\mathrm{rank}(B)\leq d}\mathbb{E}\big[\|Y - BZ\|_2^2\big]. \tag{15}$$

Next, by population least squares, $B^{\mathrm{raw}}_{\mathrm{OLS}} = \Sigma_{YZ}\Sigma_{ZZ}^{-1}$ is the unconstrained minimizer of $\mathbb{E}[\|Y - BZ\|_2^2]$. Therefore, for every $B \in \mathbb{R}^{Kd\times D_x}$,

$$\mathbb{E}\big[\|Y - BZ\|_2^2\big] = \mathbb{E}\big[\|Y - B^{\mathrm{raw}}_{\mathrm{OLS}}Z\|_2^2\big] + \left\|(B - B^{\mathrm{raw}}_{\mathrm{OLS}})\Sigma_{ZZ}^{1/2}\right\|_F^2.$$

Thus

$$\inf_{\mathrm{rank}(B)\leq d}\mathbb{E}\big[\|Y - BZ\|_2^2\big] = \mathbb{E}\big[\|Y - B^{\mathrm{raw}}_{\mathrm{OLS}}Z\|_2^2\big] + \inf_{\mathrm{rank}(B)\leq d}\left\|B\Sigma_{ZZ}^{1/2} - T_K\right\|_F^2.$$

Since $\Sigma_{ZZ}^{1/2}$ is invertible, the rank constraint remains $\mathrm{rank}(B\Sigma_{ZZ}^{1/2}) \leq d$. By the Eckart-Young theorem [17] for Frobenius-norm low-rank approximation,

$$\inf_{\mathrm{rank}(B)\leq d}\left\|B\Sigma_{ZZ}^{1/2} - T_K\right\|_F^2 = \sum_{j>d}\sigma_j(T_K)^2. \tag{16}$$

Combining this with (15) yields

$$\inf_{W,A}\mathbb{E}\big[\|Y - WAZ\|_2^2\big] = \mathbb{E}\big[\|Y - B^{\mathrm{raw}}_{\mathrm{OLS}}Z\|_2^2\big] + \sum_{j>d}\sigma_j(T_K)^2. \tag{17}$$

The claimed inequality follows immediately. □

Remark 3.11 (The capacity term). The bound $\mathcal{E}_{\text{tail}}(d) = \sum_{j>d} \sigma_j(T_K)^2$ is architecturally irreducible: no choice of (W, A) can recover the predictive content beyond the top-d singular directions of T_K. The spectral profile $\{\sigma_j(T_K)\}_{j>d}$ thus quantifies the unavoidable cost of the embedding bottleneck d.

Proposition 3.10 sharpens the saturation of Section 3.2: the embedding dimension d is an architectural ceiling that additional target blocks cannot bypass. The next subsection studies the finite-sample room for improvement within this population ceiling.

3.4 Post-Saturation Gains via Eigenvalue Balancing

By Corollary 3.6, $r_\star(K)$ eventually saturates at some value at most d. We focus on the regime in which the saturated value reaches the architectural ceiling, namely $r_\star(K) = d$. Additional targets cannot enlarge the accessible predictive subspace (Definition 2.5), but they can still improve finite-sample recovery through the eigenvalue conditioning of the whitened end-to-end predictor $T_K = B^{\text{raw}}_{\text{OLS}} \Sigma_{ZZ}^{1/2}$ from Proposition 3.10. Let $H_K := T_K^\top T_K$. To quantify this, define

$$\kappa_K^2 := \frac{\lambda_d(H_K)}{\lambda_1(H_K)} = \left(\frac{\sigma_d(T_K)}{\sigma_1(T_K)}\right)^2. \tag{18}$$

This quantity is scale-invariant and satisfies $0 < \kappa_K^2 \le 1$.

Assumption 3.12 (Finite-window post-saturation relative geometry)**.** There exist constants $K^\star \le K_{\max}$, $\kappa_0 > 0$, $\gamma > 0$, and $r_{\text{eff}} > 0$ such that for every $K \in \{K^\star, \dots, K_{\max}\}$,

$$r_\star(K) = d, \qquad \operatorname{rank}(T_K) = d, \qquad \kappa_K^2 \ge \kappa_0^2 + \gamma(K - K^\star), \qquad \operatorname{effdim}(T_K) \le r_{\text{eff}}.$$

Let $V_K, \widehat{V}_K \in \mathcal{V}_d$ denote the top-d right singular subspaces of T_K and $\widehat{T}_K$, respectively.

Proposition 3.13 (Post-saturation recovery under relative conditioning)**.** *Under Assumption 3.12, for any $K \in \{K^\star, \dots, K_{\max}\}$ and any $\delta \in (0, 1)$, with probability at least $1 - \delta$,*

$$\|\sin\Theta(V_K, \widehat{V}_K)\|_{\text{op}} \le \frac{C\sqrt{\frac{r_{\text{eff}} + \log(1/\delta)}{n}}}{\sqrt{\kappa_0^2 + \gamma(K - K^\star)} - C\sqrt{\frac{r_{\text{eff}} + \log(1/\delta)}{n}}}, \tag{19}$$

whenever the denominator is positive.

Proof. By Assumption 3.12, $\operatorname{rank}(T_K) = d$, so the Wedin gap $g_d(T_K)$ reduces to $\sigma_d(T_K)$. By Lemma A.2 and Lemma A.3 applied to T_K and $\widehat{T_K}$,

$$\|\sin\Theta(V_K, \widehat{V}_K)\|_{\text{op}} \le \frac{\epsilon_n}{\sigma_d(T_K) - \epsilon_n}, \tag{20}$$

where

$$\epsilon_n = C\|T_K\|_{\text{op}} \sqrt{\frac{\operatorname{effdim}(T_K) + \log(1/\delta)}{n}}. \tag{21}$$

Because $\sigma_d(T_K) = \kappa_K \sigma_1(T_K) = \kappa_K \|T_K\|_{\text{op}}$ by (18), we can divide numerator and denominator of (20) by $\sigma_1(T_K) > 0$, obtaining

$$\|\sin\Theta(V_K, \widehat{V}_K)\|_{\text{op}} \le \frac{C\sqrt{\frac{\operatorname{effdim}(T_K) + \log(1/\delta)}{n}}}{\kappa_K - C\sqrt{\frac{\operatorname{effdim}(T_K) + \log(1/\delta)}{n}}}.$$

Substituting $\operatorname{effdim}(T_K) \le r_{\text{eff}}$ in the numerator and $\kappa_K \ge \sqrt{\kappa_0^2 + \gamma(K - K^\star)}$ in the denominator both from Assumption 3.12 yields (19). ☐

The bound (19) couples the finite-sample term and the deterministic floor in a single ratio, obscuring the K-dependence. To expose the K-rate, we impose a *half-gap condition* requiring the finite-sample term to be at most half the relative singular-value gap:

$$C\sqrt{\frac{r_{\text{eff}} + \log(1/\delta)}{n}} \le \frac{1}{2}\sqrt{\kappa_0^2 + \gamma(K - K^\star)}. \tag{22}$$

so that the bound simplifies to an explicit rate in K as shown below. This condition is mild in n, and we verify in Section 4 that it is non-vacuous across the K-range studied.

Corollary 3.14 (Finite-window post-saturation rate)**.** *Under Assumption 3.12 and the half-gap condition* (22)*, for any* $K \in \{K^\star, \dots, K_{\max}\}$ *and any* $\delta \in (0,1)$*, with probability at least* $1-\delta$*,*

$$\|\sin\Theta(V_K, \widehat{V}_K)\|_{\mathrm{op}} \le \frac{2C}{\sqrt{\kappa_0^2 + \gamma(K-K^\star)}} \sqrt{\frac{r_{\mathrm{eff}} + \log(1/\delta)}{n}}. \tag{23}$$

Proof. Under condition (22), the denominator in (19) is at least $\frac{1}{2}\sqrt{\kappa_0^2 + \gamma(K-K^\star)}$. Substituting this lower bound into (19) gives (23). □

To translate subspace recovery into excess risk, for predictive subspace $V \in \mathcal{V}_d$, define

$$\mathcal{R}_K(V) := \frac{1}{K} \inf_{W,A:\, \mathrm{row}(WA) \subseteq V} \mathbb{E}\big[\|Y - WAZ\|^2\big], \tag{24}$$

the optimal per-target risk among predictors using V. The factor $1/K$ removes trivial growth in target dimension, so variation in $\mathcal{R}_K$ reflects predictive quality rather than aggregation. This is the normalized counterpart of the K-block risk in Proposition 3.10.

Assumption 3.15 (Subspace-Lipschitz excess risk)**.** Let $\mathcal{R}_K^\star := \mathcal{R}_K(V_K)$. There exists a constant $L > 0$ such that for every $K \in \{K^\star, \dots, K_{\max}\}$,

$$\mathcal{R}_K(\widehat{V}_K) - \mathcal{R}_K^\star \le L\, \|\sin\Theta(V_K, \widehat{V}_K)\|_{\mathrm{op}}. \tag{25}$$

Corollary 3.16 (Normalized excess-risk bound)**.** *Under Assumptions 3.12 and 3.15 and the half-gap condition* (22)*, for any* $K \in \{K^\star, \dots, K_{\max}\}$ *and any* $\delta \in (0,1)$*, with probability at least* $1-\delta$*,*

$$\mathcal{R}_K(\widehat{V}_K) - \mathcal{R}_K^\star \le \frac{2LC}{\sqrt{\kappa_0^2 + \gamma(K-K^\star)}} \sqrt{\frac{r_{\mathrm{eff}} + \log(1/\delta)}{n}}. \tag{26}$$

Proof. The claim follows by combining Corollary 3.14 with Assumption 3.15. □

Combining Corollary 3.16 with the population floor from Proposition 3.10 bounds $\mathcal{R}_K(\widehat{V}_K)$ itself.

Corollary 3.17 (Risk decomposition)**.** *Under Assumptions 3.12 and 3.15 and the half-gap condition* (22)*, for any* $K \in \{K^\star, \dots, K_{\max}\}$ *and any* $\delta \in (0,1)$*, with probability at least* $1-\delta$*,*

$$\mathcal{R}_K(\widehat{V}_K) \le \underbrace{\tfrac{1}{K}\big(\mathbb{E}\big[\|Y - B_{\mathrm{OLS}}^{\mathrm{raw}} Z\|^2\big] + \mathcal{E}_{\mathrm{tail}}(d)\big)}_{\mathcal{R}_K^\star,\ \textit{population rank-d risk}} + \underbrace{\frac{2LC}{\sqrt{\kappa_0^2 + \gamma(K-K^\star)}} \sqrt{\frac{r_{\mathrm{eff}} + \log(1/\delta)}{n}}}_{\textit{excess risk}}. \tag{27}$$

Proof. By definition $\mathcal{R}_K(\widehat{V}_K) = \mathcal{R}_K^\star + (\mathcal{R}_K(\widehat{V}_K) - \mathcal{R}_K^\star)$. The second term is bounded by Corollary 3.16; the first equals the underbraced architectural floor by Proposition 3.10. □

The population term $R_K^\star$ is the per-target rank-d optimum from Proposition 3.10, consisting of the unconstrained OLS risk plus the rank-d spectral tail of T_K. The excess-risk term in (27) is then governed by two finite-sample effects: shrinkage in n at fixed K (Remark 3.18), and post-saturation conditioning gains in K at fixed n (Remark 3.19).

Remark 3.18 (Sample complexity)*.* Inverting the excess risk (26), we obtain $\mathcal{R}_K(\widehat{V}_K) - \mathcal{R}_K^\star \le \varepsilon$ with probability at least $1-\delta$ requires $n = \widetilde{\mathcal{O}}\big(L^2\, r_{\mathrm{eff}} / \big(\kappa_0^2 + \gamma(K-K^\star)\big)\, \varepsilon^{-2}\big)$ samples. The rate is parametric in ε; the only lever on the prefactor is the post-saturation conditioning $\kappa_0^2 + \gamma(K-K^\star)$ of the whitened end-to-end predictor T_K, deferred to Section 3.6.

Remark 3.19 (Second saturation)*.* Corollary 3.6 establishes a population saturation of $r_\star(K)$. Even when this saturation occurs at the architectural ceiling $r_\star = d$, Corollaries 3.16 and 3.17 establish a second, finite-sample saturation: recovery continues to improve as new blocks balance the eigenstructure of H_K, until κ_K^2 stops improving, with absolute ceiling 1.

3.5 Risk Reduction in Multi-Target JEPA

The previous subsections identify two distinct mechanisms by which additional target blocks improve JEPA. Theorem 3.5 shows that new targets can increase predictive rank by enlarging the range of the predictor-side Gram matrix $G_K = \Sigma_{YX}^\top \Sigma_{YX}$. In contrast, Proposition 3.13 shows that, once $r_\star(K) = d$, additional targets can still improve recovery by balancing the spectrum of the whitened end-to-end predictor $T_K := B_{\mathrm{OLS}}^{\mathrm{raw}} \Sigma_{ZZ}^{1/2}$. Let $V_{K,r}, \widehat{V}_{K,r} \in \mathcal{V}_r$ denote the top-r right singular subspaces of T_K and $\widehat{T}_K$, respectively. Here we combine them in a single risk decomposition.

Proposition 3.20 (Unified finite-sample risk decomposition)**.** *Fix K and $r \le d$. Let $\Delta_{K,r} := \sigma_r(T_K) - \sigma_{r+1}(T_K)$, with the convention $\sigma_{r+1}(T_K) = 0$ when $r = \operatorname{rank}(T_K)$, and let $\epsilon_n(T_K) := C\|T_K\|_{\mathrm{op}}\sqrt{(\operatorname{effdim}(T_K) + \log(1/\delta))/n}$. Assume $\epsilon_n(T_K) < \Delta_{K,r}$, and suppose the event*

$$\mathcal{E} := \left\{ \|\widehat{T}_K - T_K\|_{\mathrm{op}} \le \epsilon_n(T_K) \right\}$$

holds with probability at least $1-\delta$. Then, under rank-r version of Assumption 3.15, on the event $\mathcal{E}$,

$$\mathcal{R}_K(\widehat{V}_{K,r}) \le \underbrace{\frac{1}{K}\left(\mathbb{E}\big[\|Y - B_{\mathrm{OLS}}^{\mathrm{raw}} Z\|^2\big] + \sum_{j>r} \sigma_j(T_K)^2 \right)}_{\mathcal{R}_K^\star(r),\ \text{population rank-}r\text{ risk}} + \underbrace{L\frac{\epsilon_n(T_K)}{\Delta_{K,r} - \epsilon_n(T_K)}}_{\text{excess-risk bound}}. \tag{28}$$

Proof. By the reduced-rank regression solution in Lemma 2.7, using the whitening argument as in Proposition 3.10, $\mathcal{R}_K^\star(r)$ is exactly the optimal population risk among rank-r end-to-end predictor. Next, on the event $\mathcal{E}$, the Wedin perturbation bound gives

$$\|\sin\Theta(V_{K,r}, \widehat{V}_{K,r})\|_{\mathrm{op}} \le \frac{\epsilon_n(T_K)}{\Delta_{K,r} - \epsilon_n(T_K)},$$

because $\Delta_{K,r} = \sigma_r(T_K) - \sigma_{r+1}(T_K)$ and $\epsilon_n(T_K) < \Delta_{K,r}$. This is the same perturbation step used in Proposition 3.13, now applied at rank r. By Assumption 3.15,

$$\mathcal{R}_K(\widehat{V}_{K,r}) - \mathcal{R}_K^\star(r) \le L\|\sin\Theta(V_{K,r}, \widehat{V}_{K,r})\|_{\mathrm{op}}.$$

Substituting the Wedin bound into this inequality gives (28). □

Remark 3.21 (Finite-risk improvement). The bound (28) decomposes the risk of the learned rank-r predictor, with $r \le d$, into three distinct terms. The first term is determined by the predictability of the task itself. It is irreducible once the targets and context are given, and can be monitored by the predictive rank $r_\star(K)$ (Definition 3.4) or partially by heterogeneity level (Definition 3.7). The second term is the population rank-r floor. It is capacity-controlled by architecture at most $r = d$. Thus, if $d \ge r_\star(K)$, choosing $r = r_\star(K)$ removes this truncation floor. If $d < r_\star(K)$, even the largest choice $r = d$ leaves the bottleneck tail $\frac{1}{K}\sum_{j>d}\sigma_j(T_K)^2$. The third term is the finite-sample recovery error. This term is controlled by the eigenstructure of T_K: a larger retained gap and a better balanced spectrum. It can be monitored by $\kappa_K^2 = \lambda_d(H_K)/\lambda_1(H_K)$ for $H_K = T_K^\top T_K$.

Thus, given the context and targets, the two controllable levers are capacity and conditioning. Increasing the bottleneck d reduces the population truncation term until $d = r_\star(K)$. Additionally, better eigenstructure of whitened end-to-end predictor T_K, reduces the excess risk bound.

3.6 Predictive Isotropy and Regularization

Building upon Proposition 3.20 and Remark 3.21, we now show that the eigenvalue-balancing condition of Proposition 3.13 corresponds precisely to isotropic covariance of the end-to-end prediction on its rank-d range. Furthermore, there exists a gauge-invariant factorization to select a JEPA predictor-encoder pair enforcing isotropic-Gaussian embedding (Section 3.6.1). Recall the stacked context $Z \in \mathbb{R}^{D_x}$ from Section 3.3, the linear context encoder A from Assumption 2.3, and the whitened end-to-end predictor T_K with $H_K := T_K^\top T_K$ from Proposition 3.10.

The object of interest is the end-to-end prediction $P_K := T_K\widetilde{Z}$, i.e., $P_K = B_{\mathrm{OLS}}^{\mathrm{raw}} Z$ written in whitened coordinates. The rank-d JEPA end-to-end prediction WAZ approximates P_K under the embedding bottleneck (14). Whereas LeJEPA's SIGReg regularizes the law of the embedding AZ toward an isotropic Gaussian prior [5], we target the prediction-aware analogue on P_K. We show that the second-moment quantity of this connection is already a sharp spectral condition on H_K.

Lemma 3.22 (Predictive isotropy)**.** *Let $H_K = T_K^\top T_K$. Assuming $\Sigma_{ZZ} \succ 0$, $\mathbb{E}[\widetilde{Z}] = 0$, and $\operatorname{rank}(T_K) = d$, the end-to-end prediction $P_K = T_K\widetilde{Z}$ has isotropic covariance on its rank-d predictive range if and only if*

$$\lambda_1(H_K) = \lambda_2(H_K) = \cdots = \lambda_d(H_K). \tag{29}$$

Proof. Let $T_K = U_K S_K V_K^\top$ be a compact singular value decomposition, where $U_K \in \mathbb{R}^{Kd\times d}$ and $V_K \in \mathbb{R}^{D_x\times d}$ have orthonormal columns, and $S_K = \operatorname{diag}(\sigma_1(T_K), \ldots, \sigma_d(T_K))$. Since $\operatorname{rank}(T_K) = d$, $P_K = T_K\widetilde{Z}$ lies in $\operatorname{col}(T_K) = \operatorname{col}(U_K)$. Thus $U_K^\top P_K$ is the intrinsic coordinate vector of P_K on its rank-d predictive range. Covariance isotropy of P_K on this range is therefore equivalent to covariance isotropy of $U_K^\top P_K$ on $\mathbb{R}^d$. Using $P_K = T_K \widetilde{Z}$ and $\mathrm{Cov}(\widetilde{Z}) = I_{D_x}$,

$$\mathrm{Cov}(U_K^\top P_K) = \mathrm{Cov}(U_K^\top T_K \widetilde{Z}) = \mathrm{Cov}(S_K V_K^\top \widetilde{Z}) = S_K^2.$$

Thus P_K has isotropic covariance on its rank-d range if and only if $S_K^2 = cI_d$ for some $c > 0$. The nonzero eigenvalues of $H_K = T_K^\top T_K$ are the diagonal entries of S_K^2, so this is exactly (29). □

We now show predictive isotropy (Lemma 3.22) is the unique minimizer of the excess-risk bound.

Theorem 3.23 (Predictive isotropy minimizes the excess-risk bound)**.** *Fix $\tau > 0$ and consider rank-d operators T_K satisfying $\mathrm{tr}(H_K) = \|T_K\|_F^2 = \tau$. Among all such operators, the isotropic spectrum $\lambda_1(H_K) = \cdots = \lambda_d(H_K) = \tau/d$ minimizes the excess-risk bound (Corollary 3.16):*

Proof. Let $\lambda_1 \geq \cdots \geq \lambda_d > 0$ denote the nonzero eigenvalues of H_K, with $\sum_{i=1}^d \lambda_i = \tau$. At fixed $\mathrm{tr}(H_K) = \tau$, $\lambda_d \leq \tau/d \leq \lambda_1$, with both equalities if and only if the spectrum is flat; hence the isotropic spectrum simultaneously minimizes λ_1 and maximizes $\lambda_d = \sigma_d(T_K)^2$, which uniquely maximizes $\kappa_K^2 = \lambda_d/\lambda_1 \leq 1$. Since $\|T_K\|_{\mathrm{op}}^2 = \lambda_1$ and $\mathrm{effdim}(T_K) = \sum_{i=1}^d \lambda_i/\lambda_1 = \tau/\lambda_1$, and the perturbation scale in Proposition 3.13 has the form

$$\epsilon_n(T_K) = C\|T_K\|_{\mathrm{op}} \sqrt{\frac{\mathrm{effdim}(T_K) + \log(1/\delta)}{n}},$$

we obtain

$$\epsilon_n(T_K) = C\sqrt{\frac{\tau + \lambda_1 \log(1/\delta)}{n}}.$$

Combining $\epsilon_n \downarrow$ and $\sigma_d \uparrow$, the isotropic spectrum minimizes the Wedin upper bound $\epsilon_n/(\sigma_d - \epsilon_n)$ in (19) whenever the denominator is positive. □

Remark 3.24 (Majorization principle). Theorem 3.23 is the recovery-bound instance of a broader majorization principle: at fixed signal energy $\mathrm{tr}(H_K) = \tau$, the flat spectrum $(\tau/d, \ldots, \tau/d)$ is the unique majorization-minimum among positive rank-d spectra with total mass τ [18, 19]. The same principle maximizes every Schur-concave functional of the spectrum at the flat spectrum; in particular, the Gaussian-entropy functional $\frac{d}{2}\log(2\pi e) + \frac{1}{2}\sum_{i=1}^d \log \lambda_i(H_K)$ is maximized there, by the arithmetic-geometric mean (AM-GM) inequality. The finite-sample condition $\kappa_K^2 \to 1$ is then the empirical manifestation of uniform predictive-energy allocation across the rank-d range.

3.6.1 Predictive isotropy subsumes encoder isotropy.

Lemma 3.22 admits a distributional reading that places the present subsection one step further down the JEPA pipeline than LeJEPA's SIGReg. Recall that for a measure μ and a map f, the pushforward $f_\#\mu$ denotes the law of $f(X)$ when $X \sim \mu$. SIGReg regularizes the embedding pushforward $\mathcal{L}(AZ) = A_\#\mathcal{L}(Z)$ toward an isotropic Gaussian prior [5]; the prediction-aware analogue is the end-to-end predictive pushforward

$$\mathcal{L}(P_K) = (T_K)_\#\mathcal{L}(\widetilde{Z}). \tag{30}$$

Remark 3.25 (Relation to encoder isotropic regularization). Isotropy of the predictive pushforward $\mathcal{L}(P_K)$ (30) on its rank-d range is exactly the eigenvalue-balancing condition of $H_K = T_K^\top T_K$ of Lemma 3.22, now lifted from covariance to the full law. The relation to SIGReg is sharpened in the regularization paragraph below.

Within Lemma 3.22's condition, the encoder-predictor factorization remains a structural choice. We specify one that bridges the predictive isotropy with the embedding isotropy of canonical JEPA paradigms [5]. Recall the SVD of the whitened end-to-end predictor, $T_K = U_K S_K V_K^\top$, and let $V_K^\top \widetilde{Z} \in \mathbb{R}^d$ be the *SVD coordinate* of the whitened context.

Lemma 3.26 (Gauge invariance of the rank-d optimum)**.** *Let a gauge $G : \mathbb{R}^d \to \mathbb{R}^d$ be any invertible map. Define a encoder-predictor factorization (Λ_G, F_G)*

$$\Lambda_G := G \circ V_K^\top \Sigma_{ZZ}^{-1/2}, \qquad F_G := U_K S_K \circ G^{-1}.$$

Then $F_G \circ \Lambda_G$ realizes the population rank-d end-to-end prediction P_K for every choice of G.

By Lemma 3.26, predictive isotropy $H_K = \alpha I_d$ is gauge-invariant, so it constrains the end-to-end map T_K without committing to any embedding distribution. The gauge G is therefore free to be chosen for downstream purposes. In particular, it can be selected to realize either of the two canonical encoder-side regularization paradigms in JEPA: covariance isotropy (whitening [20] or VICReg-style [6]) or Gaussian isotropy (LeJEPA [5], SIGReg-style).

Theorem 3.27 (Subsumes covariance isotropy)**.** *For any orthogonal gauge $G \in \mathbb{R}^{d\times d}$, the encoder-predictor factorization (Λ_G, F_G) from Lemma 3.26 induces an isotropic encoder covariance:*

$$\mathrm{Cov}(\Lambda_G(Z)) = I_d.$$

Proof. By Lemma 3.26, $\Lambda_G(Z) = GV_K^\top \Sigma_{ZZ}^{-1/2} Z$, so

$$\mathrm{Cov}(\Lambda_G(Z)) = GV_K^\top \Sigma_{ZZ}^{-1/2}\, \Sigma_{ZZ}\, \Sigma_{ZZ}^{-1/2} V_K G^\top = GV_K^\top V_K G^\top = GG^\top = I_d,$$

using $V_K^\top V_K = I_d$ from the SVD and $GG^\top = I_d$ by orthogonality property. □

Theorem 3.27 realizes covariance isotropy with a linear gauge. We next show that a nonlinear gauge can realize the strictly stronger property of isotropic-Gaussian embedding under a mild assumption.

Theorem 3.28 (Subsumes Gaussian isotropy)**.** *Assume $V_K^\top \widetilde{Z}$ has absolutely continuous law on $\mathbb{R}^d$. Then there exists an invertible gauge $G^\star : \mathbb{R}^d \to \mathbb{R}^d$ such that the encoder-predictor factorization $(\Lambda_{G^\star}, F_{G^\star})$ from Lemma 3.26 induces an isotropic Gaussian embedding:*

$$\Lambda_{G^\star}(Z) \;\sim\; \mathcal{N}(0, I_d).$$

Proof Sketch. Existence of $G^\star$ follows from Brenier's theorem [21], since the law of $V_K^\top \widetilde{Z}$ admits a density on $\mathbb{R}^d$ by hypothesis and $\mathcal{N}(0, I_d)$ admits a density on $\mathbb{R}^d$. The encoder-predictor pair $(\Lambda_{G^\star}, F_{G^\star})$ then satisfies the conclusion by construction (Lemma 3.26). □

Subsumption in Theorems 3.27 and 3.28 means the converse fails. As encoder isotropy alone does not imply predictive isotropy, since it is compatible with arbitrary predictor conditioning $\kappa_K^2 \in (0, 1]$. Predictive isotropy is therefore strictly stronger, controlling the predictive spectrum that encoder-only regularization cannot reach. Below we show they share the same excess-risk bound of Corollary 3.16.

Remark 3.29 (Gauge-invariant excess-risk bound)*.* By Lemma 3.26, every choice of G realizes the same end-to-end prediction $P_K = T_K \widetilde{Z}$, and hence the same optimality condition as Theorem 3.23. So selecting a gauge in Theorems 3.27 and 3.28 costs nothing against the excess-risk bound.

Corollary 3.30 (Linear-Gaussian special case)**.** *If the context follow a Gaussian distribution $Z \sim \mathcal{N}(0, \Sigma_{ZZ})$, the encoder-predictor factorization reduces to the linear pair*

$$\Lambda_I = V_K^\top \Sigma_{ZZ}^{-1/2}, \qquad F_I = U_K S_K.$$

Proof. The whitened context $\widetilde{Z} = \Sigma_{ZZ}^{-1/2} Z \sim \mathcal{N}(0, I_{D_x})$ under Gaussian context, and the rows of $V_K^\top$ are orthonormal, so $\Lambda_I Z = V_K^\top \widetilde{Z} \sim \mathcal{N}(0, I_d)$. Hence $G^\star = I$ realizes Theorem 3.28, and the encoder-predictor factorization specializes to (Λ_I, F_I). □

We now turn this principle into a regularizer. The excess-risk bound in Corollary 3.16 motivates isotropy of the gauge-free predictive object H_K. The subsumption results in Theorems 3.27 and 3.28 further show that, under such regularization, encoder-side isotropy need not be imposed separately, nor must a gauge in Lemma 3.26 be chosen manually.

3.6.2 Maximum-entropy regularization induces isotropic-Gaussian embedding.

The excess-risk bound prescribes only the spectral condition (29) on H_K, a second-moment constraint on the rank-d range of P_K. This places candidate regularizers on a two-axis frontier: *fidelity*, whether $\mathcal{R} = 0$ implies (29), and *commitment*, how much distributional structure $\mathcal{R} = 0$ enforces beyond it. The Pareto-optimal choice is the one that saturates fidelity with minimal commitment. At trace τ, AM-GM selects the least-committing faithful covariance $\frac{\tau}{d} I_d$; the maximum-entropy law under this covariance is $\mathcal{N}(0, \frac{\tau}{d} I_d)$, as shown in Remark 3.24.

Therefore, we propose a maximum-entropy regularizer for the intrinsic predicted output $\widehat{P}_K^{\mathrm{int}} := \widehat{U}_K^\top \widehat{W} \widehat{A} Z \in \mathbb{R}^d$, where $\widehat{U}_K \in \mathbb{R}^{Kd\times d}$ spans the learned rank-d predictive range. By the Cramér–Wold theorem [22], in its hyperspherical form [5, Lemma 3], the target law $\widehat{P}_K^{\mathrm{int}} \sim \mathcal{N}(0, I_d)$ is equivalent to $u^\top \widehat{P}_K^{\mathrm{int}} \sim \mathcal{N}(0,1)$ for every $u \in \mathbb{S}^{d-1}$. We approximate this all-direction condition by sampling $|\mathbb{A}|$ random unit vectors $\mathbb{A} \subset \mathbb{S}^{d-1}$, projecting the predictor output onto each direction, and averaging the standard-normal Gaussianity statistics T (Epps–Pulley [23]):

$$\mathcal{R}_{\mathrm{PI}}(\widehat{P}_K^{\mathrm{int}}; \mathbb{A}) := \frac{1}{|\mathbb{A}|} \sum_{u \in \mathbb{A}} T\big(\{u^\top \widehat{p}_n\}_{n=1}^N\big)\,, \tag{31}$$

where $\widehat{p}_n := \widehat{U}_K^\top \widehat{W} \widehat{A} z_n$ is the intrinsic predicted output on the n-th sample.

By Lemma 3.22, the covariance part of this target law $\mathcal{N}(0, I_d)$ recovers the predictive-isotropy condition on the nonzero spectrum of H_K; the Gaussian part is the maximum-entropy completion of

Experiment	Sweep	Fixed quantities	Main outcome	Figure
K-saturation	K	$d = r_\star$	target-count saturation	Figure 1
Heterogeneity	$\widehat{\lambda}_{\rm het}$	$K = 32,\ d = 16$	finite-sample recovery	Figure 2
Overlap	overlap ω	$K = 24,\ d = 16$	redundancy failure mode	Figure 3
Bottleneck	d	rich target bank	spectral-tail floor	Figure 4
Post-saturation	K	$r_\star(K) = d$	relative conditioning	Figure 5
Predictive isotropy	spectrum of H_K	rank, trace, K	fixed-trace isotropy	Figure 6
Embed./pred. isotropy	spectra	rank, trace, K	isotropy distinction	Figure 9

Table 1: Synthetic experiment suite. All experiments use the same frozen linear-Gaussian teacher family; only the indicated sweep variable changes.

that covariance constraint. (31) is the predictive analogue of LeJEPA's SIGReg [5], applied to the predictor output rather than the encoder embedding. Where SIGReg constrains $\mathcal{L}(AZ)$ on the input side, $\mathcal{R}_{\rm PI}$ constrains $\mathcal{L}(\widehat{P}_K^{\rm int})$ on the output side, the distributional object whose covariance spectrum is tied to the excess-risk bound through $H_K = T_K^\top T_K$ (Theorem 3.23).

We regularize the gauge-free predictive object H_K toward isotropy to minimize the excess-risk bound in Corollary 3.16. In distributional form, this applies the maximum-entropy principle to the intrinsic predictive signal $P_K^{\rm int}$ without manually choosing a gauge in Lemma 3.26. Together with Theorems 3.27 and 3.28, this shows that isotropic-Gaussian embeddings are encoder-side realizations of the same predictive-isotropy principle. This justifies and unifies the two encoder-side regularization paradigms in canonical JEPA [20, 6, 5] as factorized instances of the predictive-isotropy condition for empirical risk minimization.

4 Experimental Studies

Setup. We evaluate the theory in a controlled frozen-teacher linear-Gaussian setting. Each example is generated from a latent state $s \sim \mathcal{N}(0, I_{r_\star})$, a context view $z = Cs + \sigma_z \epsilon_z$, and K target embeddings $y_k = A_k s + \sigma_y \epsilon_k$. The context operator $C \in \mathbb{R}^{D_x \times r_\star}$ is a fixed random orthonormal frame ($C^\top C = I$), so the experiments isolate the role of target structure rather than context-encoder conditioning. The stacked response is $Y = (y_1^\top, \ldots, y_K^\top)^\top$, and the relevant population operator is the stacked target-context covariance $\Sigma_{YZ} = AC^\top$ for the rank-growth diagnostics. For the bottleneck and post-saturation diagnostics, the theory-facing operator is the whitened raw predictor $T_K = \Sigma_{YZ}^{(K)} \Sigma_{ZZ}^{-1/2}$.

Models. We compare the unconstrained ordinary least-squares predictor with its rank-d reduced-rank regression (RRR) truncation. The RRR rank d plays the role of the JEPA embedding bottleneck. Because the teacher and context maps are frozen and linear, this setting directly tests the rank, conditioning, and spectral-tail mechanisms in the theory without confounding optimization or encoder dynamics.

Data. The synthetic target bank separates two mechanisms. Target heterogeneity changes the angular spread of a latent target dictionary, while target overlap changes how concentrated the target-bank coverage is over that dictionary. Unless otherwise stated, $r_\star = 16$, $D_x = 32$, and each target block is scalar. We use ten deterministic seeds and report means with standard-error bands. The suite summary in Table 1 covers the four ablations, the post-saturation diagnostic, and the fixed-trace predictive-isotropy ablation. Additional generator and configuration details are reported in Section D.

Metrics. All paper-facing risk panels use the average per-target normalization of $\mathcal{R}_K$ in (24). This is especially important for the K-saturation and post-saturation sweeps, where the stacked target dimension changes with K. In the heterogeneity experiment, we additionally include a fixed-bank MSE probe on a common rich reference target bank. This probe is not the canonical $\mathcal{R}_K$ risk; it asks whether the recovered subspace transfers to a held-out complete target bank. Across the first three ablations we also report recovered effective rank and the heterogeneity/conditioning quantity $\lambda_{\rm het} = \lambda_{{\rm rank}(G)}(G)$, where $G = \Sigma_{ZY}\Sigma_{YZ}$ and has the same nonzero spectrum as the latent target Gram matrix induced by A. In the heterogeneity experiment, the horizontal axis is the measured finite-sample value $\widehat{\lambda}_{\rm het}$, rather than the generator knob, so the plot directly tests the paper's conditioning variable. Effective rank is computed by counting singular values above 0.05 of the largest singular value. Solid curves show finite-sample estimates averaged over seeds; dashed curves show the corresponding population values computed from the known synthetic operators. For the bottleneck ablation, we additionally report the model effective rank, the excess risk over OLS, and the normalized spectral-tail ratio $\sum_{j>d} \sigma_j^2 / \sum_j \sigma_j^2$ of the whitened population predictor. For the predictive-isotropy ablation, we fix both rank and total predictive energy and vary only the spectrum of $H_K = T_K^\top T_K$. We report the relative conditioning $\kappa_K^2 = \lambda_d(H_K)/\lambda_1(H_K)$ and the normalized log-determinant score

$$\frac{1}{d}\sum_{i=1}^{d} \log \lambda_i(H_K) - \log\left(\frac{1}{d}\sum_{i=1}^{d} \lambda_i(H_K)\right),$$

which equals zero for a flat spectrum and becomes more negative as predictive energy becomes anisotropic.

Baselines. The main baseline is the population counterpart of each finite-sample metric: it is computed directly from the known operators A and C, without sampling noise. For the bottleneck experiment, unconstrained OLS is the reference model, and the bottleneck spectral-tail result predicts that the excess risk of the rank-d predictor is governed by the spectral tail beyond the top d predictive directions.

Results. The results in Figure 1 support the target-count saturation claim: per-target MSE decreases and recovered rank increases as complementary targets are added, then both saturate once the available predictive rank is exhausted. The results in Figure 2 show that larger measured $\widehat{\lambda}_{\text{het}}$ improves finite-sample recovery. The canonical per-target risk $\mathcal{R}_K$ remains nearly flat because increasing heterogeneity makes the target bank less redundant and therefore changes the task being averaged. By contrast, fixed-bank MSE decreases because the learned subspace transfers better to a common rich target bank; recovered rank rises and the latent predictive structure becomes identifiable. The results in Figure 3 give the complementary failure mode: when target coverage becomes redundant, increasing target overlap reduces effective rank and worsens per-target MSE. Finally, Figure 4 is consistent with the bottleneck prediction: per-target MSE decreases with d, the learned model rank grows as $\min\{d, r_\star\}$, and the finite-sample excess-risk ratio tracks both the plug-in spectral-tail ratio and the population spectral-tail ratio. Together, these results are consistent with the paper's central message: multi-target prediction helps when targets expose complementary predictive structure, but target redundancy and embedding bottlenecks limit the attainable representation. The diagnostic in Figure 5 further separates rank growth from post-saturation conditioning. Here the target bank is already full rank from $K^\star = d$ onward, so additional targets cannot increase $r_\star(K)$. Instead, targets allocated to weak predictive directions improve $\kappa_K^2 = \lambda_d(H_K)/\lambda_1(H_K)$ for $H_K = T_K^\top T_K$ and reduce finite-sample error over a finite window, while effective dimension, half-gap, and per-target risk-stability diagnostics remain controlled. Panel (d) directly compares the per-target excess-risk left-hand side with the empirical $\widehat{L}\|\sin\Theta_T\|_{\text{op}}$ upper curve from the subspace-Lipschitz excess-risk condition, thereby exposing the recovery term used by the post-saturation corollary. The calibrated concentration constant in (21) is $\widehat{C}_{0.9} = 1.49$ for $\delta = 0.1$; panels (e) and (f) use this constant with $\widehat{L}$ to plot the calibrated excess-risk and risk-decomposition bounds from Corollaries 3.16 and 3.17. The half-gap condition in (22) is empirically non-vacuous: with $\widehat{\eta}_K = \|\widehat{T}_K - T_K\|_{\text{op}}/\|T_K\|_{\text{op}}$, the condition $\widehat{\eta}_K \le \kappa_K/2$ holds for all deterministic runs in the plotted sweep. At the weakest endpoint, $K = 12$, we observe $\widehat{\eta}_{\text{mean}} = 0.0469$, $\widehat{\eta}_{\max} = 0.0543$, and $\kappa_K/2 = 0.0707$. Appendix Section D gives the exact target-bank construction and additional half-gap numerics.

Finally, Figure 6 tests the operator-level Gaussianity prediction in the new predictive-isotropy section. In this experiment, $r_\star = d$, $K = d$, and $\operatorname{tr}(H_K)$ are fixed across all conditions; only the eigenvalue profile of H_K changes. This isolates predictive spectral allocation from rank growth and total signal strength. A flat spectrum maximizes the entropy score and κ_K^2, while geometric spectral decay makes the predictive signal increasingly anisotropic on its rank-d range. The observed $\|\sin\Theta_T\|_{\text{op}}$ and common-bank excess MSE both increase as the spectrum becomes less isotropic, consistent with the finite-sample recovery mechanism. The training-bank per-target MSE is nearly flat because the experiment holds total target energy fixed; this is the intended control, not a failure mode. It shows that the recovery change is driven by spectral balance rather than by more rank or more predictive energy. Appendix Figure 9 further separates embedding-level isotropy from predictive isotropy by varying an embedding covariance spectrum independently from the predictive spectrum. In that diagnostic, common-bank excess risk is essentially unchanged by the embedding isotropy score once the predictive spectrum is fixed, but changes by orders of magnitude with $\kappa_K^2(T_K)$. This supports

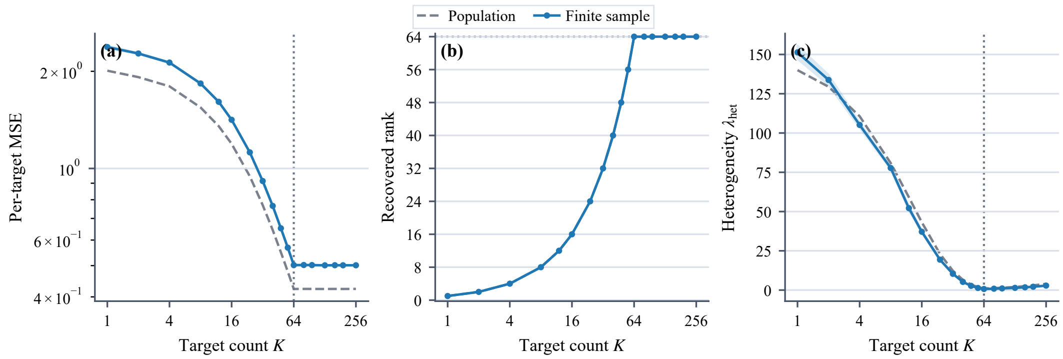

Figure 1: K-saturation under a fixed embedding bottleneck. Increasing the number of complementary target blocks lowers average per-target MSE and reveals additional predictive directions until the available predictive rank saturates.

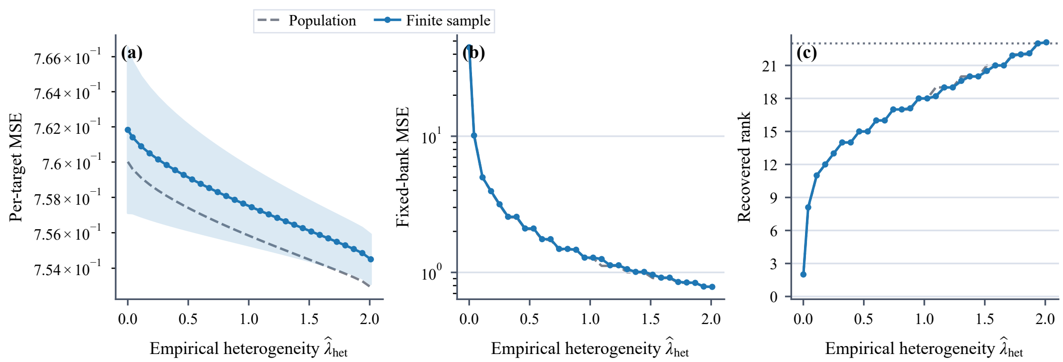

Figure 2: Effect of target heterogeneity. Larger measured $\widehat{\lambda}_{\text{het}}$ reveals additional effective predictive rank. The per-target risk panel reports $\mathcal{R}_K$ on the changing target bank, while the fixed-bank panel probes transfer to a common rich target bank.

the interpretation that embedding isotropy is at best a proxy, while the theory-facing quantity is the operator-level isotropy of H_K.

5 Discussion and Conclusion

To be filled.

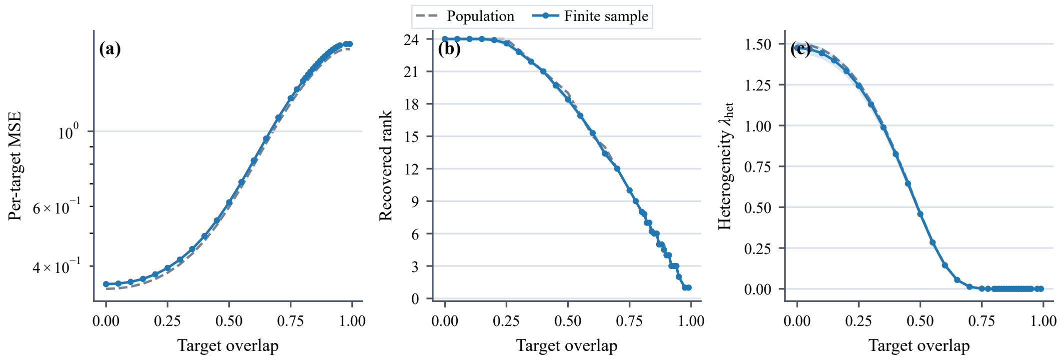

Figure 3: Effect of target overlap. Redundant targets reduce the heterogeneity/effective-rank gain expected from larger multi-target prediction objectives and worsen average per-target MSE.

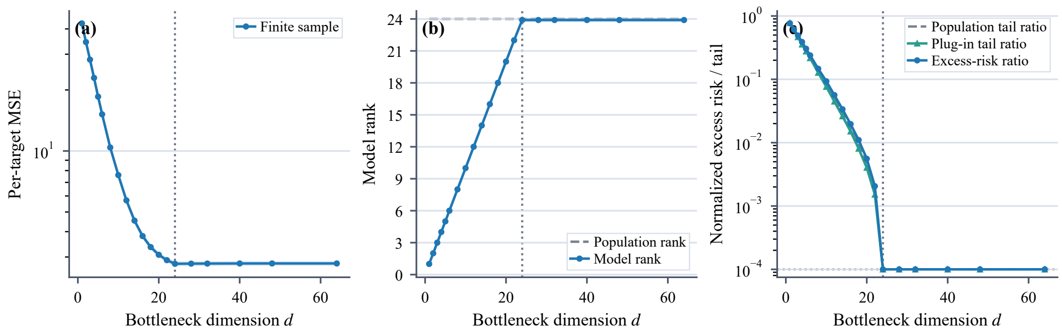

Figure 4: Embedding bottleneck ablation. The target bank has latent predictive rank $r_\star = 16$ while the allowed bottleneck rank d is swept. Per-target MSE decreases as d exposes more predictive directions. The model rank panel is expected to follow $\min\{d, r_\star\}$, while the normalized excess risk tracks the spectral-tail ratio and vanishes once $d \geq r_\star$.

Impact Statement

By the theoretical nature of this work, we do not anticipate any negative social impact.

References

[1] Yann LeCun et al. A path towards autonomous machine intelligence version 0.9. 2, 2022-06-27. *Open Review*, 62(1):1–62, 2022.

[2] Mahmoud Assran, Quentin Duval, Ishan Misra, Piotr Bojanowski, Pascal Vincent, Michael Rabbat, Yann LeCun, and Nicolas Ballas. Self-supervised learning from images with a joint-embedding predictive architecture. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 15619–15629, 2023.

[3] Adrien Bardes, Quentin Garrido, Jean Ponce, Xinlei Chen, Michael Rabbat, Yann LeCun, Mido Assran, and Nicolas Ballas. V-jepa: Latent video prediction for visual representation learning. 2023.

[4] Adrien Bardes, Jean Ponce, and Yann LeCun. Mc-jepa: A joint-embedding predictive architecture for self-supervised learning of motion and content features. *arXiv preprint arXiv:2307.12698*, 2023.

[5] Randall Balestriero and Yann LeCun. Lejepa: Provable and scalable self-supervised learning without the heuristics. *arXiv preprint arXiv:2511.08544*, 2025.

[6] Adrien Bardes, Jean Ponce, and Yann LeCun. Vicreg: Variance-invariance-covariance regularization for self-supervised learning. *arXiv preprint arXiv:2105.04906*, 2021.

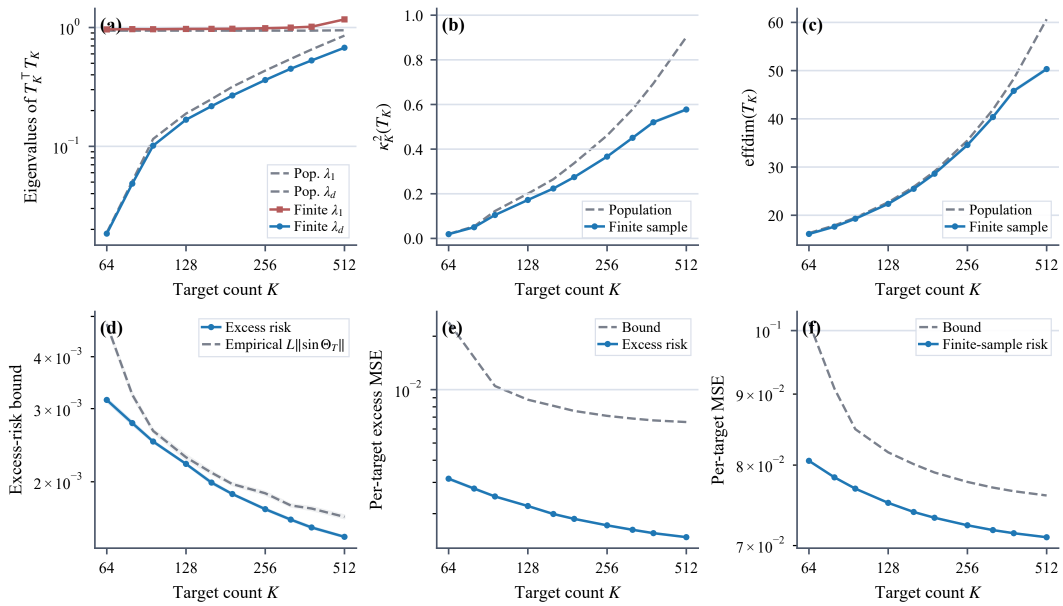

Figure 5: Post-saturation relative-conditioning validation. The target bank is full rank from $K^\star = d$ onward, so $r_\star(K) = d$ throughout. Additional targets are allocated to weak predictive directions, increasing $\kappa_K^2 = \lambda_d(H_K)/\lambda_1(H_K)$ for $H_K = T_K^\top T_K$ over a finite window. The first row reports eigenvalues of H_K, relative conditioning, and effective dimension of T_K; the second row reports the risk-stability bound, excess risk, and finite-sample risk. In panel (d), the solid curve is the empirical excess risk $\widehat{\mathcal{R}}_K - \mathcal{R}_K^\star$, while the dashed curve is $\widehat{L}\|\sin\Theta_T(V_K, \widehat{V}_K)\|_{\mathrm{op}}$. Here $\widehat{L}$ is the maximum of $[\widehat{\mathcal{R}}_K - \mathcal{R}_K^\star]_+/\|\sin\Theta_T(V_K, \widehat{V}_K)\|_{\mathrm{op}}$ over all deterministic seeds and all plotted K. Panel (e) compares the per-target excess risk with the calibrated excess-risk bound in Corollary 3.16. Panel (f) compares the finite-sample per-target risk $\widehat{\mathcal{R}}_K$ with the calibrated risk-decomposition bound in Corollary 3.17. The half-gap condition in (22) is non-vacuous and holds in all deterministic runs in the plotted sweep.

[7] Jean-Bastien Grill, Florian Strub, Florian Altché, Corentin Tallec, Pierre H Richemond, Elena Buchatskaya, Carl Doersch, Bernardo Avila Pires, Zhaohan Daniel Guo, Mohammad Gheshlaghi Azar, et al. Bootstrap your own latent: A new approach to self-supervised learning. In *Advances in Neural Information Processing Systems*, 2020.

[8] Mathilde Caron, Hugo Touvron, Ishan Misra, Hervé Jégou, Julien Mairal, Piotr Bojanowski, and Armand Joulin. Emerging properties in self-supervised vision transformers. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 9650–9660, 2021.

[9] Ting Chen, Simon Kornblith, Mohammad Norouzi, and Geoffrey Hinton. A simple framework for contrastive learning of visual representations. In *International conference on machine learning*, pages 1597–1607. PmLR, 2020.

[10] Li Jing, Pascal Vincent, Yann LeCun, and Yuandong Tian. Understanding dimensional collapse in contrastive self-supervised learning. In *International Conference on Learning Representations (ICLR)*, 2022.

[11] Yuandong Tian, Xinlei Chen, and Surya Ganguli. Understanding self-supervised learning dynamics without contrastive pairs. In *International Conference on Machine Learning*, pages 10268–10278. PMLR, 2021.

[12] Sofiane Mansouri et al. Understanding early collapse in predictive world-model pretraining. *OpenReview*, 2026.

[13] Randall Balestriero, Nicolas Ballas, Mike Rabbat, and Yann LeCun. Gaussian embeddings: How jepas secretly learn your data density. *arXiv preprint arXiv:2510.05949*, 2025.

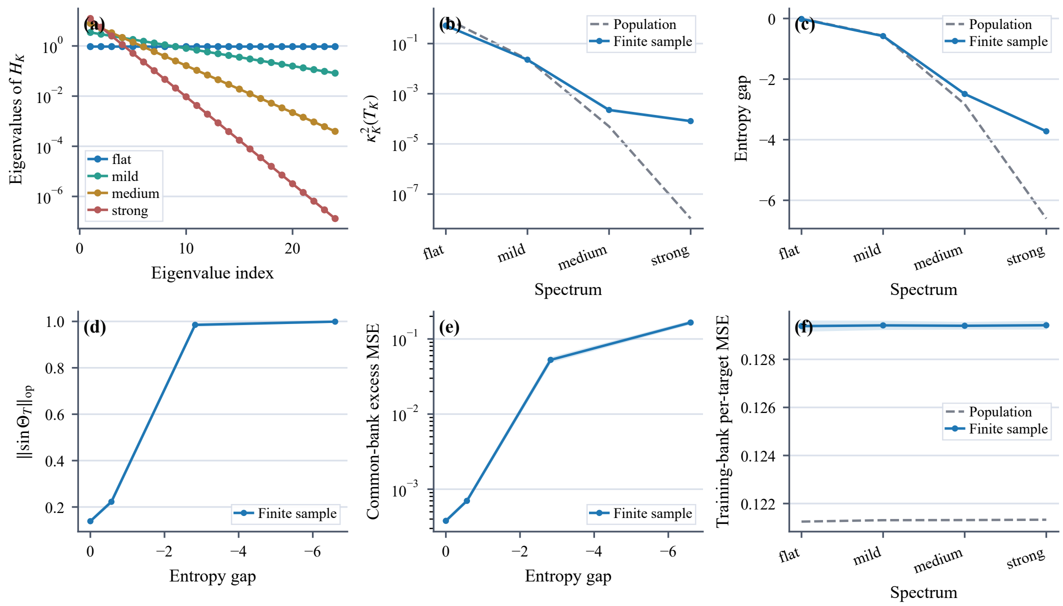

Figure 6: Predictive isotropy at fixed rank and fixed trace. All conditions have $r_\star = d = 16$, $K = 16$, and the same $\operatorname{tr}(H_K)$ for $H_K = T_K^\top T_K$; only the eigenvalue profile changes. A flatter spectrum gives larger κ_K^2, a larger log-determinant entropy score, smaller $\|\sin\Theta_T\|_{\text{op}}$, and lower excess MSE on a common flat evaluation bank. The training-bank per-target MSE remains nearly unchanged because the total predictive energy is fixed by construction.

[14] Theodore Wilbur Anderson. Estimating linear restrictions on regression coefficients for multivariate normal distributions. *The Annals of Mathematical Statistics*, pages 327–351, 1951.

[15] Alan Julian Izenman. Reduced-rank regression for the multivariate linear model. *Journal of multivariate analysis*, 5(2):248–264, 1975.

[16] Gregory C Reinsel and Raja P Velu. *Multivariate reduced-rank regression*. Springer, 1998.

[17] Carl Eckart and Gale Young. The approximation of one matrix by another of lower rank. *Psychometrika*, 1(3):211–218, 1936.

[18] A Wayne Roberts and Dale E Varberg. *Convex Functions: Convex Functions*, volume 57. Academic Press, 1974.

[19] Albert W Marshall, Ingram Olkin, and Barry C Arnold. Inequalities: theory of majorization and its applications. 1979.

[20] Aleksandr Ermolov, Aliaksandr Siarohin, Enver Sangineto, and Nicu Sebe. Whitening for self-supervised representation learning. In *ICML*, 2021.

[21] Cédric Villani et al. *Optimal transport: old and new*, volume 338. Springer, 2009.

[22] Harald Cramér and Herman Wold. Some theorems on distribution functions. *Journal of the London Mathematical Society*, s1-11(4):290–294, 1936.

[23] T. W. Epps and Lawrence B. Pulley. A test for normality based on the empirical characteristic function. *Biometrika*, 70(3):723–726, 1983.

[24] Joel A Tropp. An introduction to matrix concentration inequalities. *Foundations and trends® in machine learning*, 8(1-2):1–230, 2015.

[25] Vladimir Koltchinskii and Karim Lounici. Concentration inequalities and moment bounds for sample covariance operators. *Bernoulli*, pages 110–133, 2017.

[26] Per-Åke Wedin. Perturbation bounds in connection with singular value decomposition. *BIT Numerical Mathematics*, 12(1):99–111, 1972.

Appendix

A Supplementary Theoretical Backgrounds **20**

B Proofs of Main Text **20**

C Additional Numerical Experiments **20**

D Additional Experimental Details **20**

A Supplementary Theoretical Backgrounds

Lemma A.1 (Cross-covariance concentration [24]). *Let $(X_i, Y_i)_{i=1}^n$ be i.i.d. samples with $X \in \mathbb{R}^d$ and $Y \in \mathbb{R}^{Kd}$, and assume X and Y are sub-Gaussian with bounded second moments. Let $\widehat{\Sigma}_{YX} := \frac{1}{n}\sum_{i=1}^n Y_i X_i^\top$ denote the empirical cross-covariance. Then for any $\delta \in (0,1)$, with probability at least $1-\delta$,*

$$\|\widehat{\Sigma}_{YX} - \Sigma_{YX}\|_{\mathrm{op}} \le C\|\Sigma_{YX}\|_{\mathrm{op}}\sqrt{\frac{\mathrm{effdim}(\Sigma_{YX}) + \log(1/\delta)}{n}},$$

where $C > 0$ is an absolute constant and $\mathrm{effdim}(\Sigma_{YX}) := \frac{\|\Sigma_{YX}\|_F^2}{\|\Sigma_{YX}\|_{\mathrm{op}}^2}$ is the effective rank of Σ_{YX}.

Proof sketch. Apply the matrix Bernstein inequality [24] to the i.i.d. sum $\widehat{\Sigma}_{YX} - \Sigma_{YX} = \frac{1}{n}\sum_{i=1}^n (Y_i X_i^\top - \Sigma_{YX})$ of centered random matrices, with variance proxy controlled by the sub-Gaussian norms of X and Y. Replacing the ambient dimension Kd with the intrinsic effective rank $\mathrm{effdim}(\Sigma_{YX})$ follows from the refinement of [25]. □

Lemma A.2 (Concentration of the whitened raw predictor). *Let $T_K := B_{\mathrm{OLS}}^{\mathrm{raw}}\Sigma_{ZZ}^{1/2} = \Sigma_{YZ}\Sigma_{ZZ}^{-1/2}$ and $\widehat{T}_K := \widehat{\Sigma}_{YZ}\widehat{\Sigma}_{ZZ}^{-1/2}$. Under the same sub-Gaussian assumptions as Lemma A.1, and assuming $\lambda_{\min}(\Sigma_{ZZ}) > 0$, there exists a constant $C > 0$ such that, for any $\delta \in (0,1)$, with probability at least $1-\delta$,*

$$\|\widehat{T}_K - T_K\|_{\mathrm{op}} \le C\|T_K\|_{\mathrm{op}}\sqrt{\frac{\mathrm{effdim}(T_K) + \log(1/\delta)}{n}}. \tag{32}$$

Proof. Decompose

$$\widehat{T}_K - T_K = (\widehat{\Sigma}_{YZ} - \Sigma_{YZ})\widehat{\Sigma}_{ZZ}^{-1/2} + \Sigma_{YZ}\big(\widehat{\Sigma}_{ZZ}^{-1/2} - \Sigma_{ZZ}^{-1/2}\big).$$

The first term is controlled by the sub-Gaussian cross-covariance concentration bound from Lemma A.1 and the event $\|\widehat{\Sigma}_{ZZ}^{-1/2}\|_{\mathrm{op}} \le C\|\Sigma_{ZZ}^{-1/2}\|_{\mathrm{op}}$. For the second term, standard perturbation bounds for the inverse square root, together with sub-Gaussian concentration of $\widehat{\Sigma}_{ZZ}$, give

$$\|\widehat{\Sigma}_{ZZ}^{-1/2} - \Sigma_{ZZ}^{-1/2}\|_{\mathrm{op}} \le C\|\Sigma_{ZZ}^{-1/2}\|_{\mathrm{op}}\sqrt{\frac{\mathrm{effdim}(T_K) + \log(1/\delta)}{n}}.$$

Combining the two bounds by the triangle inequality and using $T_K = \Sigma_{YZ}\Sigma_{ZZ}^{-1/2}$ yields (32). □

Lemma A.3 (Wedin-type subspace perturbation [26]). *Let $M \in \mathbb{R}^{m\times p}$, let $r \le \min(m,p)$, and let $\widehat{M}$ be a perturbation with $\|\widehat{M} - M\|_{\mathrm{op}} \le \epsilon$. Denote by $V_r, \widehat{V}_r$ the top-r right singular subspaces of M and $\widehat{M}$, and define the* Wedin gap at level r *as $g_r(M) := \sigma_r(M) - \sigma_{r+1}(M)$, with the convention $\sigma_{r+1}(M) := 0$ when $\mathrm{rank}(M) \le r$. Provided $\epsilon < g_r(M)$,*

$$\|\sin\Theta(V_r, \widehat{V}_r)\|_{\mathrm{op}} \le \frac{\epsilon}{g_r(M) - \epsilon}.$$

Proof. Apply Wedin's $\sin\Theta$ theorem [26] to M and $\widehat{M}$ with spectral gap $g_r(M)$. □

B Proofs of Main Text

C Additional Numerical Experiments

D Additional Experimental Details

Synthetic generator. Each synthetic instance is generated from a latent linear–Gaussian model. We sample a latent vector $s \in \mathbb{R}^{r_\star}$ and observe

$$z = Cs + \sigma_z\epsilon_z, \qquad y_k = a_k^\top s + \sigma_y\epsilon_k,$$

where $C \in \mathbb{R}^{D_x \times r_\star}$ is a randomly sampled orthonormal context operator, a_k is the kth target direction, and the noise terms are independent isotropic Gaussians. Thus the population predictive structure is known, but the context coordinates are not aligned with the canonical basis.

Target banks. The target operators are constructed in latent space. In the saturation and bottleneck experiments, we use a complementary target bank whose directions progressively cover the $r_\star$ latent predictive coordinates. In the heterogeneity experiment, the construction parameter is chosen so that the induced population heterogeneity level λ_{het} is sampled approximately evenly; the plot therefore uses the measured heterogeneity level directly on the horizontal axis. In the overlap experiment, the overlap parameter concentrates target mass onto shared latent directions, reducing the effective diversity of the target bank. In the predictive-isotropy ablation, the target bank is constructed to control the spectrum directly. For a spectrum decay parameter $\rho \in (0, 1]$, we set

$$A^\top A = Q \operatorname{diag}(\lambda_1, \dots, \lambda_d) Q^\top, \qquad \lambda_i = \tau \frac{\rho^{i-1}}{\sum_{j=1}^{d} \rho^{j-1}},$$

where Q is a random orthogonal rotation and τ is fixed across all conditions. Thus $\operatorname{rank}(A) = d$ and $\operatorname{tr}(A^\top A) = \tau$ are unchanged, while the predictive spectrum ranges from flat ($\rho = 1$) to strongly anisotropic. Since the context operator has orthonormal columns and σ_z is fixed, this fixes the nonzero spectrum of $H_K = T_K^\top T_K$ up to the same scalar context-noise factor across all conditions.

Models and estimators. We first estimate the unrestricted linear predictor by ordinary least squares and then compute its rank-d reduced-rank regression approximation by truncating the singular spectrum. The learned JEPA linear predictor is therefore the best rank-constrained linear map induced by the finite training sample. Population curves use the corresponding population cross-covariance operator under the same target bank and bottleneck dimension.

Metrics. All paper-facing MSE panels report average per-target test mean-squared error, matching the normalization of $\mathcal{R}_K$ in (24). The CSV files also retain fixed-bank diagnostics on a common rich evaluation target bank as auxiliary representation probes, but these are not the main risk quantities used in the theory-facing panels. In Figure 2, we show both quantities because they answer different questions: per-target $\mathcal{R}_K$ evaluates the changing target bank used for training, whereas fixed-bank MSE evaluates the recovered subspace on a common rich target bank. As heterogeneity increases, $\mathcal{R}_K$ can remain nearly constant or rise slightly because the task becomes less redundant; the fixed-bank probe can decrease at the same time because the recovered subspace becomes more complete. The standard-error band of $\mathcal{R}_K$ also shrinks as target coverage spreads across more latent directions, so the per-target average becomes less dominated by a single shared direction. In Figure 6, panel (e) uses a related common-bank probe: after estimating the top-d right singular subspace of the whitened operator $\widehat{T}_K$, we fit a linear predictor from that recovered subspace to a fixed flat evaluation target bank. This common-bank excess MSE is not a new training objective; it is a representation diagnostic showing whether the recovered predictive subspace captures all latent directions equally well. Panel (f) keeps the canonical training-bank per-target MSE to show that total predictive energy is controlled. For the embedding-versus-predictive isotropy diagnostic, we additionally assign a synthetic embedding covariance spectrum with eigenvalues proportional to $\rho_{\mathrm{emb}}^{i-1}$ and fixed trace. This embedding spectrum is varied independently of the predictive spectrum of H_K. It should therefore be read as a controlled proxy for embedding-level isotropy diagnostics, not as a second training objective. Recovered rank is the effective numerical rank of the learned predictor, computed using a relative singular-value threshold of 0.05 times the leading singular value. The heterogeneity metric λ_{het} is the $r_\star$th nonzero eigenvalue of the target-induced Gram operator. In the latent coordinates this is the same nonzero spectrum as $A^\top A$; in the observed context coordinates it is the same nonzero spectrum as $\Sigma_{ZY}\Sigma_{YZ}$ because $C^\top C = I$. Subspace error is measured by the largest principal-angle sine between the recovered and true predictive subspaces. For the heterogeneity recovery bound, we also compute the empirical perturbation $\widehat{\epsilon} = \|\widehat{\Sigma}_{YZ} - \Sigma_{YZ}\|_{\mathrm{op}}$ and the corresponding Davis–Kahan/Wedin-style denominator $\sqrt{\lambda_{\mathrm{het}}} - \widehat{\epsilon}$. When this denominator is nonpositive, the finite-sample bound is vacuous. We keep these quantities in the CSV as validity diagnostics, but omit them from Figure 2 because the bound is loose for this sweep and is less informative than the rank and risk panels. The sharper recovery-bound visualization appears in the post-saturation experiment, where the relative conditioning condition is non-vacuous over the main finite window.

For the bottleneck experiment, the plotted spectral-tail quantity is the reduced-rank approximation error predicted by the bottleneck spectral-tail result:

$$\mathcal{E}_{\mathrm{tail}}(d) = \sum_{j>d} \sigma_j^2,$$

where $\{\sigma_j\}$ are singular values of the relevant whitened regression operator. In the figure we report the normalized ratio $\mathcal{E}_{\mathrm{tail}}(d)/\sum_j \sigma_j^2$ and compare it with the correspondingly normalized observed excess risk of the rank-d estimator over the unrestricted OLS estimator.

The rank panel in the bottleneck figure reports the effective rank of the learned reduced-rank predictor, not the rank of the stacked cross-covariance. The latter is fixed by the target bank and therefore does not change when the bottleneck dimension is swept. Since the synthetic target bank has

Table 2: Phase-1 synthetic experiment configurations.

Experiment	n_{train}	n_{test}	$r_\star$	K	d	Noise
K saturation	256	8192	16	sweep	16	0.12
Heterogeneity	32768	8192	16	32	16	0.10
Overlap	1024	8192	16	24	16	0.08
Bottleneck	96	8192	16	32	sweep	0.16
Post-saturation	5000	8192	12	sweep	12	0.25
Predictive isotropy	512	8192	16	16	16	0.25
Embedding vs. predictive isotropy	512	8192	16	16	16	0.25

latent predictive rank $r_\star = 16$, a well-estimated rank-d predictor is expected to use approximately $\min\{d, r_\star\}$ directions. This panel is therefore a sanity check that the bottleneck cap is the active rank constraint before $d = r_\star$ and that additional capacity is unused afterward.

Baselines. The main finite-sample baseline is unrestricted OLS, which removes the embedding bottleneck. Population curves are included as reference predictions for the same synthetic distribution. The bottleneck experiment additionally reports a plug-in spectral tail computed from the finite-sample OLS spectrum, allowing the risk decomposition to be checked using learned empirical quantities rather than only population quantities.

Configuration summary. We list the main Phase-1 configurations used for the reported figures in Table 2.

Additional theory diagnostics. The CSV schema also records the effective rank of the learned RRR coefficient, the finite-sample target value $\min\{d, \widehat{r}_\star(K)\}$ predicted by the rank identity, and their difference. This directly checks the rank identity beyond the displayed rank of the stacked cross-covariance.

The diagnostic in Figure 5 tests the post-saturation relative-conditioning mechanism. The target bank is full rank from $K^\star = d = 12$ onward, so additional targets do not change $r_\star(K)$. The relevant post-saturation conditioning ratio is

$$\kappa_K^2 = \frac{\lambda_d(H_K)}{\lambda_1(H_K)}, \qquad H_K = T_K^\top T_K, \quad T_K = \Sigma_{YZ}^{(K)} \Sigma_{ZZ}^{-1/2}.$$

The initial saturated target bank is deliberately imbalanced. Writing $A_K^\top A_K$ for the latent target-bank design Gram, we initialize

$$A_{K^\star}^\top A_{K^\star} = Q \operatorname{diag}(c_1, \ldots, c_{12}) Q^\top,$$

where Q is a fixed random orthogonal rotation and

$$(c_1, \ldots, c_{12}) \approx (1.00, 0.70, 0.49, 0.34, 0.24, 0.17, 0.12, 0.083, 0.058, 0.041, 0.029, 0.020).$$

Thus the bank is already rank-12, but $\kappa_{K^\star}^2 \approx 0.02$. For $K > K^\star$, each additional target row is added to the currently weakest eigendirection and contributes 0.1 units of target energy. This keeps $r_\star(K) = d$ throughout the plotted window while increasing κ_K^2. The finite-sample subspace error and per-target excess MSE decrease over the same window, supporting the repaired post-saturation claim that extra targets help only when they improve relative conditioning of the already-accessible predictive subspace. In Figure 5, we keep the core post-saturation diagnostics in the first row: the weakest and top eigenvalues of $H_K = T_K^\top T_K$ in a shared panel, the relative conditioning ratio κ_K^2, and the effective dimension of T_K. The second row reports the risk-stability bound, excess risk, and finite-sample risk. The effective-dimension panel checks that $\operatorname{effdim}(T_K)$ remains bounded over the finite window in the finite-window post-saturation relative-geometry condition. To instantiate the risk-stability constant, we compute the empirical ratio

$$\widehat{L}_K = \frac{[\widehat{\mathcal{R}}_K - \mathcal{R}_K^\star]_+}{\|\sin\Theta(V_K, \widehat{V}_K)\|_{\text{op}}},$$

using per-target excess MSE in the numerator, and set $\widehat{L} = \max_{K,\text{seed}} \widehat{L}_K$. Panel (d) then plots the excess-risk left-hand side against the empirical bound curve $\widehat{L}\|\sin\Theta(V_K, \widehat{V}_K)\|_{\text{op}}$, matching the form of the normalized risk-stability condition. Each seed is one deterministic draw of the synthetic data and target bank, and the same $\widehat{L}$ is used across the full plotted window. In the current run this gives $\widehat{L} = 6.94 \times 10^{-3}$.

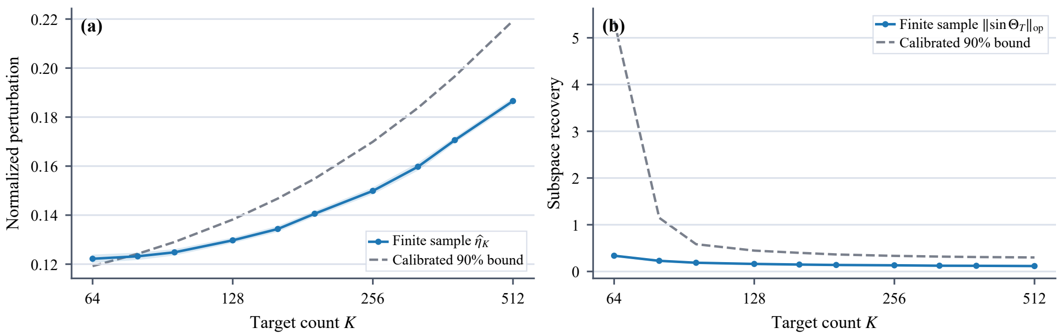

Figure 7: Empirical calibration of the concentration constant in ϵ_n from the post-saturation recovery proof. Panel (a) plots the normalized perturbation $\widehat{\eta}_K = \|\widehat{T}_K - T_K\|_{\mathrm{op}}/\|T_K\|_{\mathrm{op}}$ against the calibrated concentration bound $\widehat{C}_{0.9}\sqrt{(\mathrm{effdim}(T_K) + \log(1/\delta))/n}$. Panel (b) substitutes the same $\widehat{C}_{0.9}$ into the Wedin-style subspace-recovery bound and compares it with $\|\sin\Theta_T(V_K, \widehat{V}_K)\|_{\mathrm{op}}$.

For the concentration constant in (21), we set $\delta = 0.1$ and compute

$$a_K = \sqrt{\frac{\mathrm{effdim}(T_K) + \log(1/\delta)}{n}}, \qquad \widehat{\eta}_K = \frac{\|\widehat{T}_K - T_K\|_{\mathrm{op}}}{\|T_K\|_{\mathrm{op}}}.$$

We then set $\widehat{C}_{0.9}$ to the 90th percentile, over all plotted (K, seed) pairs, of $\widehat{\eta}_K/a_K$. This gives $\widehat{C}_{0.9} = 1.49$ and covers $90/100$ normalized perturbation samples by construction. Panel (e) uses the calibrated bound

$$\frac{2\widehat{L}\widehat{C}_{0.9}a_K}{\kappa_K}$$

from Corollary 3.16. Panel (f) adds this term to the population OLS risk estimate, matching the risk-decomposition bound in Corollary 3.17. In this post-saturation setting $d = r_\star(K)$, so the spectral-tail term is zero up to numerical precision.

To check that the simplified rate is not vacuous in the plotted regime, we evaluate the empirical analogue of the half-gap condition in (22). Define the normalized perturbation

$$\widehat{\eta}_K := \frac{\|\widehat{T}_K - T_K\|_{\mathrm{op}}}{\|T_K\|_{\mathrm{op}}}.$$

The condition $\widehat{\eta}_K \le \kappa_K/2$ holds on the seed-averaged curve for every plotted K, and it holds seed-wise for all $100/100$ deterministic runs. At the weakest endpoint $K = 12$, where the target bank is deliberately closest to the small-conditioning boundary, the whitened diagnostic still remains below the half-gap threshold in all seeds. Representative values are:

K	$\widehat{\eta}_{K,\mathrm{mean}}$	$\widehat{\eta}_{K,\mathrm{max}}$	$\kappa_K/2$	valid seeds
12	0.0469	0.0543	0.0707	10/10
16	0.0489	0.0545	0.1440	10/10
96	0.0713	0.0808	0.4771	10/10

As an appendix-only diagnostic, we also substitute the same $\widehat{C}_{0.9}$ into the intermediate Wedin-style recovery bound from the proof of Corollary 3.14. This yields the dashed subspace-recovery curve in Figure 7. The induced curve is conservative: it covers all observed $\|\sin\Theta_T\|_{\mathrm{op}}$ values in the plotted window, because the deterministic perturbation step is looser than the empirical perturbation quantile.

Predictive-isotropy construction. The fixed-trace predictive-isotropy experiment is designed to test the operator-level Gaussianity claim without changing rank or total signal energy. All conditions use $D_x = 32$, $r_\star = d = K = 16$, $\sigma_z = \sigma_y = 0.25$, and $\mathrm{tr}(H_K) = 15.06$ after the common whitening/context-noise scaling. The decay values are

$$\rho \in \{1.00,\ 0.85,\ 0.65,\ 0.45\}.$$

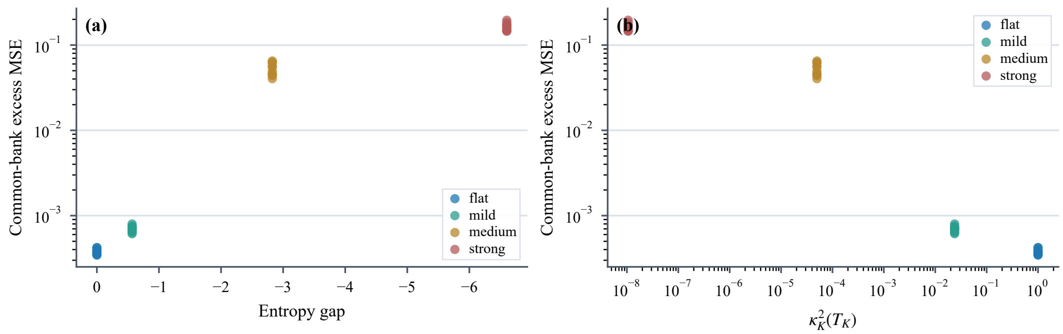

Figure 8: Seed-level predictive-isotropy scatter. Each point is one deterministic seed and one fixed-trace spectrum condition. The common-bank excess MSE rises as the predictive spectrum becomes less isotropic, whether isotropy is measured by the log-determinant entropy gap or by $\kappa_K^2(T_K)$.

They induce the following population diagnostics:

ρ	$\kappa_K^2(T_K)$	entropy gap	$\|\sin\Theta_T\|_{\mathrm{op}}$	common excess MSE
1.00	1.0000	0.000	0.136	2.96×10^{-4}
0.85	8.74×10^{-2}	-0.266	0.159	3.74×10^{-4}
0.65	1.56×10^{-3}	-1.507	0.497	2.79×10^{-3}
0.45	6.27×10^{-6}	-3.814	0.988	9.14×10^{-2}

The trace column is omitted because it is constant by construction. The strong anisotropy settings intentionally violate the half-gap condition; they are included to show the transition from the finite-window perturbative regime to the nearly unrecoverable weak-direction regime. The flat and mild settings satisfy the half-gap diagnostic in all seeds for the current sample size.

Figure 8 reports the same experiment as a seed-level scatter. The points lie on the expected monotone trend: more negative entropy gap, or smaller $\kappa_K^2(T_K)$, corresponds to larger common-bank excess MSE.

Embedding versus predictive isotropy. The embedding-versus-predictive diagnostic varies two spectra independently: the synthetic embedding covariance spectrum and the predictive spectrum of $H_K = T_K^\top T_K$. The embedding decay values are $\rho_{\mathrm{emb}} \in \{1.00, 0.65, 0.45\}$, and the predictive decay values are $\rho_{\mathrm{pred}} \in \{1.00, 0.85, 0.65, 0.45\}$. Both spectra are normalized to the same trace within their own space. This creates cases with nearly isotropic embedding covariance but badly conditioned predictive operator, and cases with anisotropic embedding covariance but well-conditioned predictive operator.

Figure 9 reports the result. Averaging over predictive spectra, the common-bank excess MSE is unchanged across the three embedding decay values. Averaging over embedding spectra, the same excess MSE varies by orders of magnitude across predictive decay values:

ρ_{pred}	common excess MSE
1.00	2.96×10^{-4}
0.85	3.74×10^{-4}
0.65	2.79×10^{-3}
0.45	9.14×10^{-2}

The Spearman correlation between embedding entropy gap and common-bank excess MSE is 0.00 in this controlled grid, while the corresponding correlation for predictive entropy gap is -0.91. This does not claim that embedding isotropy is useless in nonlinear JEPA training; it shows that embedding isotropy is not the operator-level quantity appearing in the finite-sample recovery bound.

Reproducibility. All Phase-1 experiments are run with ten deterministic seeds. The full synthetic suite can be reproduced with

```
python -m experiments.run_all
```

followed by

```
python -m plots.plot_main.
```

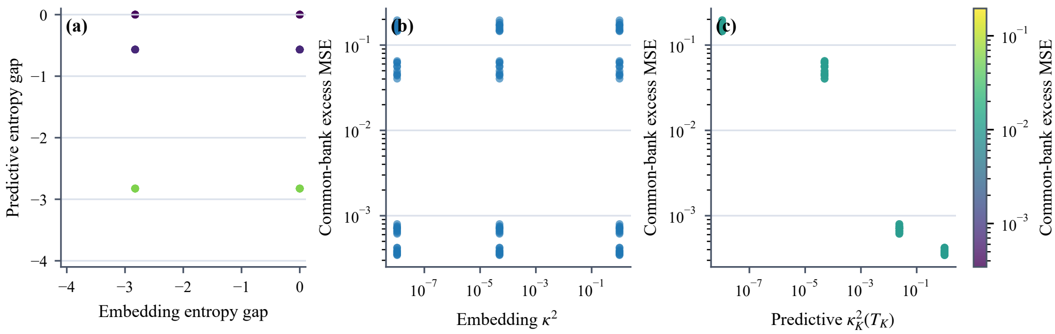

Figure 9: Embedding isotropy versus predictive isotropy. The embedding covariance spectrum and the predictive spectrum of $H_K = T_K^\top T_K$ are varied independently at fixed rank and trace. Panel (a) shows that embedding entropy and predictive entropy need not agree; color indicates common-bank excess MSE. Panel (b) shows that embedding relative conditioning alone does not explain the risk. Panel (c) shows that common-bank excess MSE follows predictive relative conditioning $\kappa_K^2(T_K)$ much more directly.

The scripts write per-experiment CSV files, the combined result table, and the main and appendix PDF figures. Unit tests cover deterministic generation, OLS/RRR behavior, rank constraints, subspace metrics, CSV schema, and figure-reference existence.

Visualization choices. Finite-sample estimates are plotted as solid curves with standard-error bands. Population predictions are plotted as dashed gray curves. For the bottleneck spectral-tail-ratio panel, values below 10^{-4} are visually floored in the log-scale display to keep the super-threshold regime legible; this affects only the display, not the CSV values or computations.

NeurIPS Paper Checklist

The checklist is designed to encourage best practices for responsible machine learning research, addressing issues of reproducibility, transparency, research ethics, and societal impact. Do not remove the checklist: **The papers not including the checklist will be desk rejected.** The checklist should follow the references and follow the (optional) supplemental material. The checklist does NOT count towards the page limit.

Please read the checklist guidelines carefully for information on how to answer these questions. For each question in the checklist:

- You should answer [Yes], [No], or [N/A].
- [N/A] means either that the question is Not Applicable for that particular paper or the relevant information is Not Available.
- Please provide a short (1–2 sentence) justification right after your answer (even for [N/A]).

The checklist answers are an integral part of your paper submission. They are visible to the reviewers, area chairs, senior area chairs, and ethics reviewers. You will also be asked to include it (after eventual revisions) with the final version of your paper, and its final version will be published with the paper.

The reviewers of your paper will be asked to use the checklist as one of the factors in their evaluation. While [Yes] is generally preferable to [No], it is perfectly acceptable to answer [No] provided a proper justification is given (e.g., error bars are not reported because it would be too computationally expensive" or "we were unable to find the license for the dataset we used"). In general, answering [No] or [N/A] is not grounds for rejection. While the questions are phrased in a binary way, we acknowledge that the true answer is often more nuanced, so please just use your best judgment and write a justification to elaborate. All supporting evidence can appear either in the main paper or the supplemental material, provided in appendix. If you answer [Yes] to a question, in the justification please point to the section(s) where related material for the question can be found.

IMPORTANT, please:

- **Delete this instruction block, but keep the section heading "NeurIPS Paper Checklist"**,
- **Keep the checklist subsection headings, questions/answers and guidelines below.**
- **Do not modify the questions and only use the provided macros for your answers**.

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: **[TODO]**

   Justification: **[TODO]**

   Guidelines:

   - The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
   - The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
   - The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
   - It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: **[TODO]**

   Justification: **[TODO]**

   Guidelines:

   - The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
   - The authors are encouraged to create a separate "Limitations" section in their paper.
   - The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.

- The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
- The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
- The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
- If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
- While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren't acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. **Theory assumptions and proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

   Answer: **[TODO]**

   Justification: **[TODO]**

   Guidelines:
   - The answer [N/A] means that the paper does not include theoretical results.
   - All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
   - All assumptions should be clearly stated or referenced in the statement of any theorems.
   - The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
   - Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
   - Theorems and Lemmas that the proof relies upon should be properly referenced.

4. **Experimental result reproducibility**

   Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

   Answer: **[TODO]**

   Justification: **[TODO]**

   Guidelines:
   - The answer [N/A] means that the paper does not include experiments.
   - If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
   - If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
   - Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general. releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
   - While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
     (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.

(b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
(c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
(d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. **Open access to data and code**

   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

   Answer: **[TODO]**

   Justification: **[TODO]**

   Guidelines:
   - The answer [N/A] means that paper does not include experiments requiring code.
   - Please see the NeurIPS code and data submission guidelines (`https://neurips.cc/public/guides/CodeSubmissionPolicy`) for more details.
   - While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
   - The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (`https://neurips.cc/public/guides/CodeSubmissionPolicy`) for more details.
   - The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
   - The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
   - At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
   - Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. **Experimental setting/details**

   Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

   Answer: **[TODO]**

   Justification: **[TODO]**

   Guidelines:
   - The answer [N/A] means that the paper does not include experiments.
   - The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
   - The full details can be provided either with the code, in appendix, or as supplemental material.

7. **Experiment statistical significance**

   Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

   Answer: **[TODO]**

   Justification: **[TODO]**

   Guidelines:
   - The answer [N/A] means that the paper does not include experiments.
   - The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
   - The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
   - The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)

- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
- If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. **Experiments compute resources**

   Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

   Answer: **[TODO]**

   Justification: **[TODO]**

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
   - The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
   - The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. **Code of ethics**

   Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics https://neurips.cc/public/EthicsGuidelines?

   Answer: **[TODO]**

   Justification: **[TODO]**

   Guidelines:

   - The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.
   - If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
   - The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. **Broader impacts**

    Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

    Answer: **[TODO]**

    Justification: **[TODO]**

    Guidelines:

    - The answer [N/A] means that there is no societal impact of the work performed.
    - If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
    - Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
    - The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
    - The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.

- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. **Safeguards**

    Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

    Answer: **[TODO]**

    Justification: **[TODO]**

    Guidelines:

    - The answer [N/A] means that the paper poses no such risks.
    - Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
    - Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
    - We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. **Licenses for existing assets**

    Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

    Answer: **[TODO]**

    Justification: **[TODO]**

    Guidelines:

    - The answer [N/A] means that the paper does not use existing assets.
    - The authors should cite the original paper that produced the code package or dataset.
    - The authors should state which version of the asset is used and, if possible, include a URL.
    - The name of the license (e.g., CC-BY 4.0) should be included for each asset.
    - For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
    - If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, `paperswithcode.com/datasets` has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
    - For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
    - If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. **New assets**

    Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

    Answer: **[TODO]**

    Justification: **[TODO]**

    Guidelines:

    - The answer [N/A] means that the paper does not release new assets.
    - Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
    - The paper should discuss whether and how consent was obtained from people whose asset is used.
    - At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. **Crowdsourcing and research with human subjects**

    Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

    Answer: **[TODO]**

    Justification: **[TODO]**

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
- According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

    Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

    Answer: **[TODO]**

    Justification: **[TODO]**

    Guidelines:

    - The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
    - Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
    - We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
    - For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.

16. **Declaration of LLM usage**

    Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

    Answer: **[TODO]**

    Justification: **[TODO]**

    Guidelines:

    - The answer [N/A] means that the core method development in this research does not involve LLMs as any important, original, or non-standard components.
    - Please refer to our LLM policy in the NeurIPS handbook for what should or should not be described.